All India Aakash Test Series for NEET - 2027

TEST – 2 (Code-A)

Click here for Code-B Sol.

Test Date: 04/10/2026

ANSWERS

1.	(4)	37.	(2)	73.	(3)	109.	(1)	145.	(4)
2.	(1)	38.	(2)	74.	(1)	110.	(1)	146.	(2)
3.	(3)	39.	(1)	75.	(1)	111.	(3)	147.	(2)
4.	(1)	40.	(2)	76.	(1)	112.	(1)	148.	(3)
5.	(1)	41.	(2)	77.	(2)	113.	(1)	149.	(1)
6.	(1)	42.	(3)	78.	(1)	114.	(2)	150.	(3)
7.	(1)	43.	(4)	79.	(4)	115.	(3)	151.	(2)
8.	(3)	44.	(1)	80.	(1)	116.	(2)	152.	(4)
9.	(2)	45.	(4)	81.	(1)	117.	(1)	153.	(3)
10.	(1)	46.	(3)	82.	(1)	118.	(3)	154.	(3)
11.	(2)	47.	(2)	83.	(3)	119.	(1)	155.	(3)
12.	(2)	48.	(1)	84.	(1)	120.	(3)	156.	(2)
13.	(4)	49.	(2)	85.	(2)	121.	(4)	157.	(1)
14.	(4)	50.	(2)	86.	(2)	122.	(1)	158.	(3)
15.	(2)	51.	(3)	87.	(4)	123.	(2)	159.	(3)
16.	(3)	52.	(1)	88.	(3)	124.	(1)	160.	(4)
17.	(1)	53.	(4)	89.	(4)	125.	(1)	161.	(2)
18.	(3)	54.	(2)	90.	(4)	126.	(1)	162.	(2)
19.	(2)	55.	(3)	91.	(3)	127.	(3)	163.	(2)
20.	(4)	56.	(3)	92.	(4)	128.	(4)	164.	(2)
21.	(1)	57.	(4)	93.	(4)	129.	(2)	165.	(1)
22.	(4)	58.	(1)	94.	(1)	130.	(3)	166.	(3)
23.	(2)	59.	(4)	95.	(3)	131.	(1)	167.	(2)
24.	(3)	60.	(2)	96.	(1)	132.	(3)	168.	(3)
25.	(4)	61.	(2)	97.	(1)	133.	(4)	169.	(3)
26.	(3)	62.	(2)	98.	(3)	134.	(2)	170.	(2)
27.	(4)	63.	(4)	99.	(1)	135.	(2)	171.	(3)
28.	(4)	64.	(4)	100.	(2)	136.	(2)	172.	(1)
29.	(2)	65.	(2)	101.	(1)	137.	(4)	173.	(2)
30.	(4)	66.	(4)	102.	(2)	138.	(3)	174.	(3)
31.	(4)	67.	(2)	103.	(1)	139.	(2)	175.	(4)
32.	(2)	68.	(1)	104.	(3)	140.	(2)	176.	(1)
33.	(4)	69.	(3)	105.	(1)	141.	(1)	177.	(3)
34.	(3)	70.	(1)	106.	(1)	142.	(3)	178.	(4)
35.	(4)	71.	(4)	107.	(1)	143.	(1)	179.	(2)
36.	(3)	72.	(4)	108.	(2)	144.	(3)	180.	(4)

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HINTS & SOLUTIONS

[PHYSICS]

1. Answer (4)

Hint: Flux, $\phi = \vec{B} \cdot \vec{A}$

Sol.: $\vec{A} = (16\ \text{m}^2)\hat{k}$

$\phi = \vec{B}.\vec{A}$

$\phi = B_0\left(13\hat{i} - 7\hat{j} + 2\hat{k}\right).\left(16\hat{k}\right)$

$\phi = 32B_0$ Wb

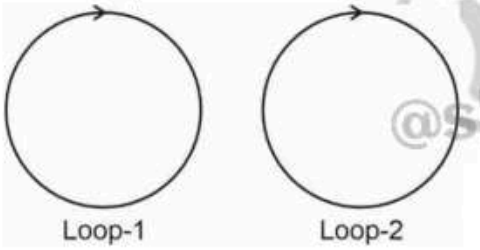
2. Answer (1)

Hint & Sol.: When magnet is either approaching or receding, it changes the flux through loop, according to Lenz's law it oppose its motion.

3. Answer (3)

Hint: Use Lenz's law.

Sol.: Let current flows in loop-1 in clockwise direction. To oppose this change current I is induced in loop-2 in clockwise direction.



$\therefore$ Loop-2 is repelled by the loop-1

4. Answer (1)

Hint: Magnitude of induced emf, $|\varepsilon| = \left|\frac{d\phi}{dt}\right|$

Sol.: $|\varepsilon| = \left|\frac{d\phi}{dt}\right| = 20$ V

5. Answer (1)

Hint: Resistance, $R = \frac{\rho l}{A}$

Sol.: $R = \frac{\rho l}{A} = \frac{\rho l^2}{V}$

$R \propto l^2 \Rightarrow \frac{100}{R'} = \frac{l^2}{4l^2} \Rightarrow R' = 400\ \Omega$

6. Answer (1)

Hint: $F = mB$

Sol.: Unit of m is A m

$B = \frac{F}{m} = \frac{\text{N.m}}{\text{A m}^2} = \frac{\text{J}}{\text{A m}^2}$

7. Answer (1)

Hint: Due to change in current in straight wire, flux is changing in rectangular loop. Use Lenz's law to find induced current in loop.

Sol.:

At time = $t_1 \rightarrow \frac{di}{dt} = +\text{ve} \Rightarrow$ Direction of induced current in anticlockwise

At time = $t_2 \rightarrow \frac{di}{dt} = -\text{ve} \Rightarrow$ Direction is clockwise

At time = $t_3 \rightarrow \frac{di}{dt} = +\text{ve} \Rightarrow$ Direction is anticlockwise

8. Answer (3)

Hint: Total charge flown

$q = \frac{\Delta\phi}{R}$ & $\Delta\phi = \vec{B} \cdot \Delta\vec{A}$

Sol.: $\Delta\phi = qR = 10^{-5} \times 20 = 2 \times 10^{-4}$ Wb

&

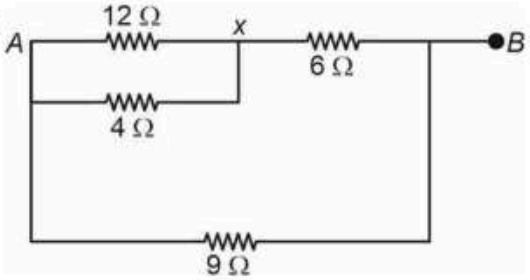
$\Delta\phi = N\left(\vec{B} \cdot \Delta\vec{A}\right) = N(B\Delta A)$ [Angle b/w B & A is 0°]

$B = \frac{\Delta\phi}{N\Delta A} = \frac{2\times10^{-4}}{20\times2\times10^{-2}} = \frac{10^{-2}}{20} = 5\times10^{-4}$ T

9. Answer (2)

Hint: Rearrange the circuit by marking potential of points

Sol.:



$R_{AB} = \frac{9}{2}\ \Omega = 4.5\ \Omega$

10. Answer (1)

Hint: Range of ammeter is given by

$\Rightarrow i = i_g\left(1 + \frac{R_G}{s}\right)$

Sol.: $i = i_g\left(1+\frac{R_G}{s}\right)$

$10 = 20\times10^{-3}\left(1+\frac{25}{s}\right)$

$\Rightarrow \quad 500 = 1+\frac{25}{s}$

$\Rightarrow \quad \frac{25}{s} = 499$

$s = \frac{25}{499}\Omega$

11. Answer (2)

Hint: For induced emf to be zero, flux should be constant.

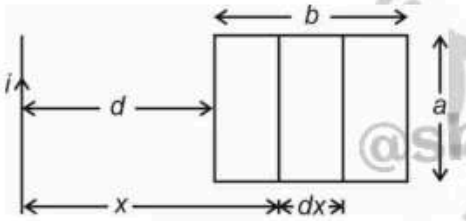
Sol.: $BlVt$ = Constant

$B = \frac{C}{lvt}$

12. Answer (2)

Hint: Mutual inductance, $M = \frac{\phi_2}{i_1}$

Sol.:



$$\int d\phi_2 = \int_d^{d+b} \frac{\mu_0 i}{2\pi x} a\,dx \Rightarrow \phi_2 = \frac{\mu_0 ia}{2\pi}\left[\ln(x)\right]_d^{d+b}$$

$$= \frac{\mu_0 ia}{2\pi}\ln\left(\frac{d+b}{d}\right)$$

$\Rightarrow \quad M = \frac{\phi_2}{i} = \frac{\mu_0(2)}{2\pi}\ln\left(\frac{5+3}{5}\right)$

$\Rightarrow \quad M = \frac{\mu_0}{\pi}\ln\left(\frac{8}{5}\right)$

13. Answer (4)

Hint: Magnetic field at a distance r at perpendicular bisector is, $B = \frac{-\mu_0\vec{M}}{4\pi r^3}$

Sol.: $B_{net} = 2\left(\frac{\mu_0 M}{4\pi\left(\frac{a}{\sqrt{2}}\right)^3}\right) = \frac{4\sqrt{2}\mu_0 M}{4\pi a^3} = \frac{\sqrt{2}\mu_0 M}{\pi a^3}$

14. Answer (4)

Hint & Sol.: For diamagnetic material, χ_m is independent of temperature.

15. Answer (2)

Hint: $\vec{F}_{net} = q\vec{E} + q\left(\vec{V}\times\vec{B}\right)$

Sol.: $\vec{F}_E = q\vec{E} = 2\times10^{-9}\times2 = \left(4\times10^{-9}\hat{k}\right)$ N

$\vec{F}_B = q(v\cos60^\circ)B\,\hat{k} = 2\times10^{-9}\times2\times\frac{1}{2}\times2$

$= \left(4\times10^{-9}\hat{k}\right)$ N.

$\vec{a} = \frac{\vec{F}_{net}}{m} = \frac{8\times10^{-9}\hat{k}}{10^{-6}} = \left(8\times10^{-3}\hat{k}\right)$ ms^{-2}

$= 0.008$ ms^{-2} $\hat{k}$

16. Answer (3)

Hint: Only magnetic force $F_m = qV_0B_0$ acts as a centripetal force

Sol.: $F_m = qV_0B_0$

$= qV_0\left(\frac{mV_0}{ql_0}\right) = \frac{mv_0^2}{l_0}$

$\therefore$ Spring remains in its natural length

$\therefore$ Potential energy remains constant

$\Rightarrow$ Since only F_m is acting $\Rightarrow$ Kinetic energy remains constant

$\Rightarrow$ F_m acts as a central force or pass through O

Therefore, Angular momentum about point O remains conserved.

17. Answer (1)

Hint & Sol.: For current through internal resistance to be maximum, equivalent resistance should be minimum, *i.e.*, $x = 0$.

18. Answer (3)

Hint & Sol.: Resistance, $R = \frac{\rho l}{A}$

It depends on dimensions of resistor.

19. Answer (2)

Hint: At null point, it will behave as balanced wheatstone bridge

Sol.: $\frac{7}{25} = \frac{R}{75}$

$\Rightarrow$ $R = 21\ \Omega$

20. Answer (4)

Hint & Sol.:

a. The intensity of magnetisation (I) is very small, negative and is directly proportional to magnetising field (H) for diamagnetic substance.

b. In non-uniform magnetic field, the paramagnetic substances are attracted towards the stronger parts of magnetic field.

c. Substances, which at room temperature, retain their ferromagnetic property for a long period of time are called permanent magnets.

21. Answer (1)

Hint: $\varepsilon_{ind} = \frac{-d\phi}{dt}$

Sol.: At $0 < t < \frac{l}{v}$

Flux is increasing downwards, so emf induced will be anticlockwise

At $\frac{l}{v} < t < \frac{3l}{v}$

Flux is constant

So, induced emf is zero.

At $\frac{3l}{v} < t < \frac{4l}{v}$

Flux is decreasing downwards, so emf induced will be clockwise.

At $t > \frac{4l}{v}$

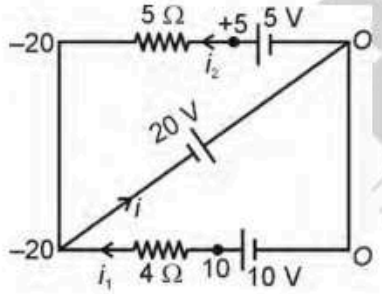
Flux is zero, so no emf will be induced.

Also, 'v' is uniform, so induced emf in the wire will be constant.

22. Answer (4)

Hint: Use Kirchhoff's law

Sol.:



$i_1 = \frac{30}{4} = 7.5$ A

$i_2 = \frac{25}{5} = 5$ A

$i = i_1 + i_2$

= 7.5 + 5

= 12.5 A

23. Answer (2)

Hint: Charge transferred, $q = \int i dt$

Sol.: $q = \int_0^2 (2 + 4t) dt$

$= 2(t)_0^2 + 2\left(t^2\right)_0^2$

= 4 + 8 = 12 C

24. Answer (3)

Hint: Energy stored across capacitor, $U = \frac{1}{2}CV^2$

Sol.: V = 100 V

$U = \frac{1}{2}CV^2 = \frac{1}{2} \times 10 \times 10^{-6} \times 10^4$

= 5 × 10^{-2} = 50 mJ

25. Answer (4)

Hint & Sol.: Since $B > B_0$, therefore material can be paramagnetic or ferromagnetic because intensity of magnetising field is positive.

26. Answer (3)

Hint: Susceptibility, $\chi = \mu_r - 1$ and $B = \mu_r \mu_0 H$

Sol.: $\mu_r = \chi + 1$

= 550

$\phi = BA = \mu_r \mu_0 HA$

= 550 × 4π × 10^{-7} × 10^3 × 2 × 10^{-5}

= 44π × 10^{-7} Wb = 4.4π × 10^{-6} Wb

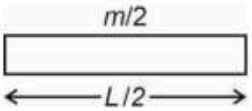
27. Answer (4)

Hint: Time period of oscillation of needle in magnetic field is given by, $T = 2\pi\sqrt{\frac{I}{MB}}$

Sol.: Initially, 

$I = \frac{mL^2}{12}$

Now



$I' = \frac{\frac{m}{2}\left(\frac{L}{2}\right)^2}{12} = \frac{m}{2} \times \frac{L^2}{4 \times 12} = \frac{I}{8}$

$M' = \frac{M}{2}$

$T' = 2\pi\sqrt{\frac{I}{8 \times \frac{M}{2} \times B}} = \frac{T}{2}$

T' = 2 s

28. Answer (4)

Hint: For paramagnetic material, $I \propto \frac{H}{T}$

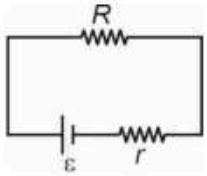
Sol.: $B = \mu H$

$$\frac{I_2}{I_1} = \frac{B_2}{B_1} \times \frac{T_1}{T_2} = \frac{1}{2} \times \frac{10}{20} = \frac{1}{4}$$

$$I_2 = \frac{I_1}{4} = \frac{0.6}{4} = 0.15 \text{ A/m}$$

29. Answer (2)

Hint: When cell is connected through external resistance R, as shown, then



$$i = \frac{\varepsilon}{R + r}$$

Sol.: $i = \frac{\varepsilon}{R + r}$

$$1 = \frac{\varepsilon}{4 + r} \Rightarrow \varepsilon = 4 + r \quad \ldots(i)$$

Now, $\frac{1}{2} = \frac{\varepsilon}{10 + r}$

$\Rightarrow \frac{10 + r}{2} = 4 + r$ [From equation (i)]

$\Rightarrow 10 + r = 8 + 2r$

$\Rightarrow r = 2\ \Omega$

30. Answer (4)

Hint: As i changes in same resistor, charge density (n) remains constant.

Sol.: $i = JA \Rightarrow J$ becomes double

$i = neAvd \Rightarrow V_d$ becomes double

31. Answer (4)

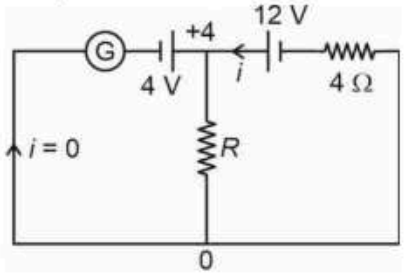
Hint and Sol.: Bismuth (Bi), copper (Cu) and Salt (NaCl) are diamagnetic substances.

Copper chloride ($CuCl_2$) is paramagnetic substance.

32. Answer (2)

Hint: Use, Kirrchhoff's law.

Sol.:



$iR = 4$ ...(1)

$\Rightarrow 12 - 4i = 4$

$\Rightarrow 4i = 8$

$\Rightarrow i = 2$ A

$\Rightarrow$ Put value in equation (1)

$R = 2\ \Omega$

33. Answer (4)

Hint & Sol.: Potential difference across a battery is more than its emf when battery is charging

34. Answer (3)

Hint: Magnetic field due to semi infinite wire, is $\frac{\mu_0 i}{4\pi d}$ and it is zero on its own axis

Sol.: $\vec{B}_p = \frac{\mu_0 i}{4\pi r} \otimes$

35. Answer (4)

Hint: Use concept of Ampere's law

Sol.: In Ampere's law, $\oint \vec{B} \cdot d\vec{l} = \mu_0 i_{enc}$, B is due to all the current present inside and outside the loop.

36. Answer (3)

Hint: This system behaves like solenoid and magnetic field inside long solenoid is given by $\mu_0 ni$

Sol.: $B = \mu_0 ni$

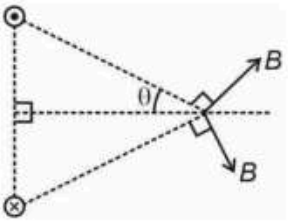
$$= 4\pi \times 10^{-7} \times \frac{100}{2\pi \times 10 \times 10^{-3}} \times 2$$

$$= 4 \times 10^{-3} \text{ T} = 4 \text{ mT}$$

37. Answer (2)

Hint: Magnetic field due to long wire at a distance r is given by, $B = \frac{\mu_0 i}{2\pi r}$

Sol.:



$\vec{B}_{net} = 2B \sin\theta$

$$= \frac{2 \times \mu_0 i}{2\pi(4)} \times \frac{1}{4}$$

$$= \frac{4\pi \times 10^{-7} \times 2}{16\pi}$$

$= 5 \times 10^{-8}$ T $= 0.05\ \mu$T

38. Answer (2)

Hint: Use Ampere's law

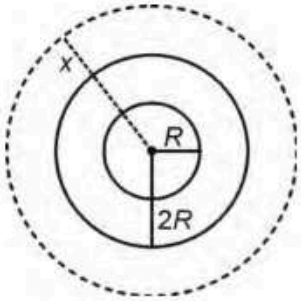
Sol.: Magnetic field due to hollow metallic pipe,

$B = 0$ for $r < R$

$$B = \frac{\mu_0 i}{2\pi r} \text{ for } r > R$$

39. Answer (1)

Hint: Use Ampere's law,



Sol.:

$$i_{enc} = \int J dA$$

$$= \int_R^{2R} J_0 r^2 . 2\pi r dr$$

$$= J_0 2\pi \int_R^{2R} r^3 dr$$

$$= J_0 2\pi \left[\frac{r^4}{4}\right]_R^{2R}$$

$$= \frac{J_0\pi}{2}\left[16R^4 - R^4\right].$$

$$i_{enc} = \frac{J_0\pi \times 15R^4}{2}$$

Now,

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 i_{enc}$$

$$B2\pi x = \frac{\mu_0 \times 15}{2} J_0 \pi R^4$$

$$B = \frac{15}{4x}\mu_0 J_0 R^4$$

40. Answer (2)

Hint: Brightness is proportional to power across lamp.

Sol.: When switch is closed, R_{eq} decrease

$\Rightarrow$ Current increases

$\therefore$ Power across A increases & across B decreases.

41. Answer (2)

Hint: V_{ab} = Potential difference across 5 V battery

$= 5 - i$

Sol.: $5 - i = 0 \Rightarrow i = 5$ A

$$\Rightarrow \quad i = \frac{25}{3+R} = 5$$

$\Rightarrow 5 = 3 + R$

$R = 2\ \Omega$

42. Answer (3)

Hint: Net force on particle is given by, $\vec{F}_{net} = q\left(\vec{v}\times\vec{B}\right) + q\vec{E}$

Sol.: Since particle is initially at rest, it will accelerate in the direction of electric field which is parallel to magnetic field. Therefore, magnetic force on particle is zero. The locus of particle will be straight line.

43. Answer (4)

Hint & Sol.: Ratio of magnitude of magnetic moment to that of its angular momentum of orbiting charge is $\frac{q}{2m}$.

44. Answer (1)

Hint: For net force to be zero in both electric and magnetic field, $\vec{E} \| \left(\vec{B}\times\vec{V}\right)$

Sol.: $\vec{E} \| \left(\vec{B}\times\vec{V}\right)$

$\therefore \quad \hat{i} \| \left(\vec{B}\times\hat{j}\right)$

$\therefore$ Direction of magnetic field should be along + z axis

45. Answer (4)

Hint: In uniform magnetic field, when particle is thrown at an angle with magnetic field, then path of particle is helical with uniform pitch and

Pitch = $v_{\|}T$ and $R = \frac{mv_{\perp}}{qB}$

Sol.: $v_{\|} = v\cos\theta$ & $v_{\perp} = v\sin\theta$

$$a = \frac{\cos 37^\circ}{\cos 53^\circ} = \frac{4\times 5}{5\times 3} = \frac{4}{3}$$

$$b = \frac{\sin 37^\circ}{\sin 53^\circ} = \frac{3}{5}\times\frac{5}{4} = \frac{3}{4}$$

$$\therefore \quad a + b = \frac{4}{3} + \frac{3}{4} = \frac{16+9}{12} = \frac{25}{12}$$

$$a - b = \frac{4}{3} - \frac{3}{4} = \frac{16-9}{12} = \frac{7}{12}$$

$$a \times b = \frac{4}{3}\times\frac{3}{4} = 1$$

[CHEMISTRY]

46. Answer (3)

Hint: The complex species which contain d electron(s) show colour due to d-d transition.

Sol.: $[Co(H_2O)_6]^{3+}$ is diamagnetic in nature and is an inner orbital complex.

$[Ti(H_2O)_6]^{3+}$ contain one d electron. It is violet in colour due to d-d transition.

47. Answer (2)

Hint: $[PtCl_2(en)_2]^{2+}$ shows geometrical isomerism

Sol.: trans-$[PtCl_2(en)_2]^{2+}$ has plane of symmetry and is optically inactive.

cis-$[PtCl_2(en)_2]^{2+}$ exist as a pair of enantiomers. Each enantiomer is chiral and optically active in nature.

48. Answer (1)

Hint: Spin only magnetic moment

$(\mu_s) = \sqrt{n(n+2)}$ BM

n = Number of unpaired electron(s)

Sol.:

Ion	Number of unpaired electron(s)	μ(BM)
Fe^{3+}	5	5.92
Co^{2+}	3	3.87
Cr^{2+}	4	4.90
Ti^{3+}	1	1.73

49. Answer (2)

Hint: More is the effective nuclear charge more will be the hydration of ions and more will be the energy released.

Sol.:

Ions		$\Delta_{hyd}H°(M^{2+})$(kJ mol^{-1})
V^{2+}	–	1895
Cr^{2+}	–	1925
Co^{2+}	–	2079
Ni^{2+}	–	2121

50. Answer (2)

Hint: Highest oxide of Mn is Mn_2O_7 while highest Mn fluoride is MnF_4.

Sol.: V_2O_5 is amphoteric in nature, hence it reacts with both acid and alkali.

CrO is basic in nature while Cr_2O_3 is amphoteric.

51. Answer (3)

Hint: $2KMnO_4 \xrightarrow{513\ K} K_2MnO_4 + MnO_2 + O_2$

Sol.: $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

52. Answer (1)

Hint and Sol.: Correct order of ligand field strength of the given species is

$C_2O_4^{2-} > F^- > S^{2-} > I^-$

53. Answer (4)

Hint: For first order reactions $kt = 2.303\log\frac{[R]_0}{[R]}$

Sol.: $k = \frac{2.303}{t}\log\frac{0.20}{0.0125}$

or, $k = \frac{2.303}{20\times60}\log16$

or, $k = \frac{2.303}{20\times60}\log(2)^4$

or, $k = \frac{2.303\times4}{20\times60}\log2$

or, $k = 2.3 \times 10^{-3}\ s^{-1}$

$t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{2.3\times10^{-3}}s = 301.3\ s$

54. Answer (2)

Hint: Oder is applicable to elementary as well as complex reactions.

Sol.: Order of a reaction is experimental quantity. It can be zero and even a fraction but molecularity cannot be zero or a non-integer.

55. Answer (3)

Hint: For zero order reaction

$\text{rate} = -\frac{1}{3}\frac{d[A]}{dt} = k[A]^0$

Sol.: $[A]_0 - [A] = 3kt$

or, $t = \frac{1}{3k}\{[A]_0 - [A]\}$

or, $t = \frac{1}{3\times2\times10^{-2}}\{2.4 - 0.48\}$

or, $t = \frac{1.92}{3\times2\times10^{-2}} = 32\ s$

56. Answer (3)

Hint: $\log\frac{k_2}{k_1}=\frac{E_a}{2.303R}\left[\frac{T_2-T_1}{T_1T_2}\right]$

Sol.: Rate of reaction α rate constant of reaction.

$$\log\frac{4}{1}=\frac{E_a}{2.303R}\left[\frac{320-300}{320\times300}\right]$$

$$\text{or, } E_a=\frac{2.303R\times320\times300}{20}\times\log4$$

$$\text{or, } E_a=\frac{2.303\times8.314\times320\times300}{20}\times2\log2$$

or, $E_a = 55.1\ kJ\ mol^{-1}$

57. Answer (4)

Hint: rate of reaction

$$=-\frac{d[A]}{dt}=-\frac{1}{3}\frac{d[B]}{dt}=\frac{1}{2}\frac{d[C]}{dt}$$

Sol.: $$-\frac{1}{3}\frac{d[B]}{dt}=\frac{1}{2}\frac{d[C]}{dt}$$

$$\text{or, } -\frac{d[B]}{dt}=\frac{3}{2}\frac{d[C]}{dt}$$

$$\text{or, } -\frac{d[B]}{dt}=\frac{3}{2}\times4\times10^{-3}\ molL^{-1}s^{-1}$$

$$-\frac{d[B]}{dt}=6\times10^{-3}\ molL^{-1}s^{-1}$$

58. Answer (1)

Hint: General electronic configuration of lanthanoids is: $[Xe]4f^{1\text{-}14}\ 5d^{0\text{-}1}6s^2$

Sol.:

Ions		Electronic configuration	Number of unpaired electrons
Er^{2+}(Z = 68)	:	$4f^{12}$	2
Tm^{2+} (Z = 69)	:	$4f^{13}$	1
Nd^{2+} (Z = 60)	:	$4f^{4}$	4
Dy^{2+} (Z = 66)	:	$4f^{10}$	4

59. Answer (4)

Hint: More is the ligand field strength more is the value of crystal field splitting energy; more will be the frequency of light absorbed and lesser is the value of wavelength of light absorbed.

Sol.: Ligand field strength order:

en > NH_3 > H_2O > F^-

Wavelength of light absorbed:

$[CoF_6]^{3-} > [Co(H_2O)_6]^{3+} > [Co(NH_3)_6]^{3+} > [Co(en)_3]^{3+}$

60. Answer (2)

Hint: Potassium dichromate is orange coloured compound

Sol.:

Species		Colour
$K_2Cr_2O_7$	–	Orange
Mn_2O_7	–	Green
MnO_4^-	–	Purple
CrO_4^{2-}	–	Yellow

61. Answer (2)

Hint: In acidic medium dichromate is reduced to Cr^{3+} and Fe^{2+} is oxidised to Fe^{3+}

Sol.:

$$Cr_2O_7^{2-}+14H^++6Fe^{2+}\rightarrow2Cr^{3+}+6Fe^{3+}+7H_2O$$

62. Answer (2)

Hint: $\ln k=\ln A-\frac{E_a}{RT}$

Sol.: $$\ln e^3=5-\frac{7.47\times10^3}{8.3\times T}$$

$$\text{or, } 3=5-\frac{0.9\times10^3}{T}$$

$$\text{or, } \frac{0.9\times10^3}{T}=2$$

$$\text{or, } T=\frac{0.9\times10^3}{2}=450\ K$$

63. Answer (4)

Hint: A Catalyst catalyses the forward as well as the backward reactions to the same extent so that the equilibrium state remains same but is reached faster.

Sol.: A catalyst catalyses the spontaneous reaction but does not catalyse non-spontaneous reactions.

64. Answer (4)

Hint: Rate = $[A_2]^x[B_2]^y$

overall order of reaction x + y

Sol.: $B_2 \rightleftharpoons B+B$

$$K=\frac{[B][B]}{[B_2]}=\frac{[B]^2}{[B_2]}$$

$$\text{or, } [B]=K^{\frac{1}{2}}[B_2]^{\frac{1}{2}}$$

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Rate of the reaction is determined by the slow step.

$$\text{Rate} = K[A_2][B] = k[A_2]K^{\frac{1}{2}}[B_2]^{\frac{1}{2}}$$

$$\text{Rate} = = kK^{\frac{1}{2}}[A_2][B_2]^{\frac{1}{2}}$$

$$\text{overall order of reaction} = 1+\frac{1}{2}=1.5$$

65. Answer (2)

Hint: $k=\frac{2.303}{t_2-t_1}\log\frac{r_1}{r_2}$ for a first order reaction

Sol.: $k=\frac{2.303}{60}\log\frac{0.08}{0.04}$

$k = 1.15 \times 10^{-2}\ s^{-1}$

66. Answer (4)

Hint: Inversion of cane sugar takes place in excess of water in acidic medium.

Sol.: $2NH_3(g)\xrightarrow[\text{Pt catalyst}]{1130K}N_2(g)+3H_2(g)$

Rate = $k[NH_3]^0 = k$

- All natural and artificial radioactive decay of unstable nuclei take place by first order kinetics.
- $\underset{\text{cane sugar}}{C_{12}H_{22}O_{11}}+H_2O\xrightarrow{\Delta}\underset{\text{Glucose}}{C_6H_{12}O_6}+\underset{\text{Fructose}}{C_6H_{12}O_6}$

  Rate = $k[C_{12}H_{22}O_{11}]$

$$2H_2O_2\xrightarrow[\text{alkaline medium}]{I^-}2H_2O+O_2$$

Rate = $k[H_2O_2][I^-]$

67. Answer (2)

Hint: As the atomic number increases, the effective nuclear charge of lanthanoid ion increases hence the basic strength of its hydroxide decreases.

Sol.: Correct order of atomic number of the lanthanoid ions: $Er^{3+} > Tb^{3+} > Eu^{3+} > Pm^{3+}$

correct order of basic strength

$Pm(OH)_3 > Eu(OH)_3 > Tb(OH)_3 > Er(OH)_3$

68. Answer (1)

Hint: $[MnBr_4]^{2-}$ is tetrahedral in shape and is a high spin complex.

Sol.: Tetrahedral complexes are high spin complexes hence d electrons in Mn^{2+} are unpaired, giving a spin only magnetic moment of 5 electrons.

69. Answer (3)

Hint: The ions which contain partially filled d orbitals show colour in aqueous medium due to d-d transition.

Sol.: d-d transition is not observed in metal ions containing completely filled d-orbitals or empty *d* subshell.

Aquated Zn^{2+}, Sc^{3+}, Ti^{4+} do not show colour in aqueous medium. The ions which do not contain *d* electrons, do not show colour in aqueous medium.

70. Answer (1)

Hint: For zero order reaction $t_{1/2}=\frac{[R]_0}{2k}$

Sol.: If $A \rightarrow P$ follows second order kinetics then,

$$\frac{-d[A]}{dt}=k[A]^2$$

$$\text{unit of } k=\frac{molL^{-1}s^{-1}}{mol^2L^{-2}}=mol^{-1}Ls^{-1}$$

71. Answer (4)

Hint: $[CoF_6]^{3-}$ is a high spin complex

Sol.: Crystal field stabilization energy for

$$[CoF_6]^{3-}=\left[\frac{-2}{5}\times4+\frac{3}{5}\times2\right]\Delta_0$$

$= [-1.6 + 1.2]\Delta_0$

$= -0.4\Delta_0$

72. Answer (4)

Hint: Metal present in cyanocobalamine is cobalt

Sol.: The pigment responsible for photosynthesis, chlorophyll, is a coordination compound of magnesium.

Haemoglobin, the red pigment of blood, which acts as oxygen carrier is a coordination compound of iron.

Vitamin B_{12}, cyanocobalamine, the antipernicious anaemia factor, is a coordination compound of cobalt.

Wilkinson catalyst, $[(Ph_3P)_3RhCl]$ is the coordination compound of rhodium and is used for the hydrogenation of alkenes.

73. Answer (3)

Hint: It is a first order reaction, as the unit of rate constant is s^{-1}.

$$kt = 2.303\log\frac{\left(p_A^{\circ}\right)}{\left(p_A^{t}\right)}$$

Sol.:

	$A(g)$	$\rightarrow 2B(g)$	$+ C(g)$
Initial pressure	320	0	0
Pressure after time t	$(320 - x)$	2x	x

$320 - x + 2x + x = 920$

$2x = 600$

$x = 300$

$$t = \frac{2.303}{k}\log\frac{320}{20}$$

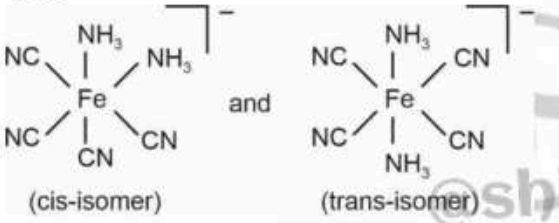
$$t = \frac{2.303}{2\times10^{-3}}\log 16$$

$$t = \frac{2.303\times4\times0.3010}{2\times10^{-3}} = 1386 \text{ s}$$

74. Answer (1)

Hint: The species which contain plane of symmetry are optically inactive.

Sol.:



(cis-isomer) (trans-isomer)

Both the isomers contain plane of symmetry hence both are optically inactive.

75. Answer (1)

Hint: More is the effective nuclear charge, more is the ionisation enthalpy

Sol.:

Element	$\Delta_i H°$ (kJ mol^{-1})
Fe	762
Mn	717
Cr	653
V	650

76. Answer (1)

Hint: Most common oxidation state of vanadium is +5

Sol.: Most common oxidation state of nickel is +2

77. Answer (2)

Hint: Tetraammineaquachloridocobalt (III) chloride is $[Co(NH_3)_4(H_2O)Cl]Cl_2$

Sol.: 1 mole of $[Co(NH_3)_4(H_2O)Cl]Cl_2$ will give 2 moles of chloride ions in aqueous medium.

Mole of AgCl precipitated would be: $200 \times 0.4 \times 2 \times 10^{-3} = 0.16$

78. Answer (1)

Hint: More is the electron density on the metal better is the back donation from metal filled *d* orbital to the vacant π^* orbital of CO.

Sol.: Correct order of C–O bond strength of the given complexes is

$[Mn(CO)_6]^+ > [Ni(CO)_4] > [Cr(CO)_6] > [Co(CO)_4]^-$

79. Answer (4)

Hint: $E°$ value of Ce^{4+}/Ce^{3+} is 1.74 V. It is a strong oxidant.

Sol.: Lanthanoid metals on reaction with dilute acid liberate hydrogen gas.

80. Answer (1)

Hint: For stable electronic configuration, the metallic bonding is weak and the metals have a lower value of enthalpy of atomisation.

Sol.:

Metals	Enthalpy of atomisation (kJ mol^{-1})
V	515
Cr	397
Mn	281
Co	425

81. Answer (1)

Hint: Interstitial compounds are those which are formed when atoms like H, C or N are trapped inside the crystal lattices of metals.

Sol.: Interstitial compounds retain metallic conductivity.

82. Answer (1)

Hint: V_2O_5 catalyses the oxidation of SO_2 in the manufacture of H_2SO_4.

Sol.: Iron catalysts are used in the Haber process for the production of NH_3 from N_2/H_2 mixtures.

cis-platin and related compounds of platinum effectively inhibit the growth of tumours.

EDTA is used in the treatment of lead poisoning.

83. Answer (3)

Hint: Complexes in which a metal is bound to only one kind of donor groups are called homoleptic complexes.

Sol.: $K_3[Cr(C_2O_4)_3]$: potassium trioxalatochromate (III) is a homoleptic complex.

84. Answer (1)

Hint: $[MnCl_6]^{3-}$ is a high spin complex.

Sol.: $[MnCl_6]^{3-}$ is an octahedral high spin complex hence electronic configuration of Mn^{3+} in the complex will be $t_{2g}^3 e_g^1$.

85. Answer (2)

Hint: In $[Ni(CN)_4]^{2-}$, hybridization of Ni is dsp^2.

Sol.:

Complex	Hybridization	Shape	Magnetic nature
$[NiCl_4]^{2-}$	sp^3	Tetrahedral	Paramagnetic
$[Ni(CO)_4]$	sp^3	Tetrahedral	Diamagnetic
$[PtCl_4]^{2-}$	dsp^2	Square planar	Diamagnetic
$[Ni(CN)_4]^{2-}$	dsp^2	Square planar	Diamagnetic

86. Answer (2)

Hint: Ligand which can ligate through two different atoms is called ambidentate ligand.

Sol.: NO_2^- ion can coordinate either through nitrogen or through oxygen to a central atom/ion.

SCN^- ion can coordinate through either the sulphur or through nitrogen atom.

87. Answer (4)

Hint: Jahn Teller effect in the coordination complexes is observed when *d* orbitals are asymmetrically filled.

Sol.: In high spin d^4, d^6 and d^7 the electrons in the d orbitals are asymmetrically filled hence the complexes with these configurations show Jahn-Teller distortion.

88. Answer (3)

Hint: $\frac{K_{(T+10)}}{K_T} = (x)^{\Delta T/10}$

x = temperature coefficient

Sol.: $\frac{K_{(T+10)}}{K_T} = (2)^{50/10}$

$r_2 = r_1(2)^5$

89. Answer (4)

Hint: Rate = $k[A]^x \times [B]^y$

Sol.: $k[0.1]^x\ [0.2]^y = 1 \times 10^{-5}$ ...(i)

$k[0.2]^x\ [0.4]^y = 8 \times 10^{-5}$ ...(ii)

$k[0.2]^x\ [0.8]^y = 32 \times 10^{-5}$ ...(iii)

Dividing (i) by (ii)

$$\left(\frac{1}{2}\right)^x \times \left(\frac{2}{4}\right)^y = \frac{1}{8}$$

$$\left(\frac{1}{2}\right)^{x+y} = \frac{1}{8} = \left(\frac{1}{2}\right)^3$$

$x + y = 3$ ...(iv)

Dividing (ii) by (iii)

$$\left(\frac{4}{8}\right)^y = \frac{8}{32}$$

$$\left(\frac{1}{2}\right)^y = \left(\frac{1}{2}\right)^2$$

$y = 2$ ...(v)

Putting the value of y in (iv) we get

$x = 3 - y = 3 - 2 = 1$

90. Answer (4)

Hint: Elements whose atomic numbers are more than that of uranium are called transuranium elements.

Sol.:

Ion	Radii (pm)
U^{3+}	103
Am^{3+}	99

[BOTANY]

91. Answer (3)

Hint: Eye colour is an X-linked character in *Drosophila*.

Sol.: Only dominant traits such as brown body colour in *Drosophila* can be expressed in both homozygous as well as heterozygous conditions.

92. Answer (4)

Hint: Accumulation of ketone bodies in the body occurs in phenylketonuria.

Sol.: Accumulation of phenylpyruvic acid in brain is observed in phenylketonuria.

93. Answer (4)

Hint: Both thalassemia and sickle-cell anaemia are recessive gene disorders.

Sol.: Thalassemia is quantitative and sickle-cell anaemia is qualitative problem related to globin molecules.

94. Answer (1)

Hint: Individuals with Turner's syndrome are sterile.

Sol.: If both parents are affected and heterozygous for myotonic dystrophy, there is a

25% chance of unaffected progeny. People affected by Down's syndrome can form gametes with normal number of autosomes.

95. Answer (3)

Hint: G. Gamow argued that the codon must be composed of a combination of nucleotides.

Sol.: George Gamow suggested that in order to code for all the 20 amino acids, the code should be made up of three nucleotides.

96. Answer (1)

Hint: Humans show XY type sex determination.

Sol.: In grasshopper (XX-XO type), maleness is determined by absence of X chromosome. In honeybee, sex is determined by overall ploidy level of the organism.

97. Answer (1)

Hint: Process C is reverse transcription.

Sol.: The relationship between template (mRNA) and product (polypeptide) of translation cannot be explained by using complementarity.

98. Answer (3)

Hint: Skin colour becomes darker with increase in number of dominant alleles present in individuals.

Sol.: Three dominant alleles in the genotype lead to intermediate skin colour. Increase in number of dominant alleles by one, would make skin colour slightly darker than the intermediate skin colour.

99. Answer (1)

Hint: An allele of each pair must be present in gametes.

Sol.: The allele Y does not exist in the genotype of the plant, so it cannot be present in the gamete as well.

100. Answer (2)

Hint: DNA replication occurs in a semiconservative manner.

Sol.: *E. coli* generation time is 20 minutes. After 2 generations (40 minutes), a heavy DNA containing cell, would produce four daughter cells, out of which two will contain light DNA and two would possess hybrid DNA.

101. Answer (1)

Hint: H. G. Khorana developed the chemical method for synthesising RNA molecules.

Sol.: The technique of DNA Fingerprinting was initially developed by Alec Jeffreys.

102. Answer (2)

Hint: Differences in DNA of members of a population at the level of 1 base, are called SNPs or Snips.

Sol.: Allelic sequence variation has traditionally been described as a DNA polymorphism if more than one variant/allele at a locus occurs in human population with a frequency greater than 0.01.

103. Answer (1)

Hint: One codon codes for one amino acid only.

Sol.: The sequence of codons in mRNA can reliably be determined only if the polypeptide is composed of non-degenerate amino acids.

104. Answer (3)

Hint: Tritiated thymidine helped Taylor to prove the DNA replication model in *Vicia faba*.

Sol.: Deoxyribose sugar is present in DNA.

105. Answer (1)

Hint: A complete turn of DNA contains 10 bp.

Sol.: If a stretch of DNA contains 150 nucleotide pair,

The total number of turns cannot exceed

$$= \frac{150}{10} = 15 \text{ turns}$$

106. Answer (1)

Hint: Prokaryotic genes are mostly polycistronic and are non-split in nature.

Sol.: A DNA sequence coding for tRNA, rRNA or polypeptide can define a gene. Most eukaryotic genes are monocistronic and are split genes. Reading frame of mRNA is finally determined post splicing.

107. Answer (1)

Hint: UGA acts as a non-sense codon.

Sol.: AUG plays an important role as the initiation codon. It also codes for methionine.

108. Answer (2)

Hint: Frederick Griffith discovered transformation.

Sol.: Avery, MacLeod and McCarty proved DNA was the transforming principle.

109. Answer (1)

Hint: Axial flower position is considered dominant.

Sol.: Since the flower position determines pod position and terminal flowers are recessive in nature, the probability of having axial green pods in the same plant post fertilisation would be 1.

110. Answer (1)

Hint: In sickle cell anemia, the mutation changes the 6th codon of mRNA from GAG to GUG.

Sol.: In sickle cell anemia, the mutation causes the Glutamic acid on the 6th position of beta globin chain to be replaced by Valine.

111. Answer (3)

Hint: The male grasshopper contains one less chromosome as compared to the female.

Sol.: Male grasshopper lacks Y chromosome. Y chromosome is found in male *Drosophila.*

112. Answer (1)

Hint: Humans and *Drosophila* are similar in terms of sex determination.

Sol.: Chickens exhibit female heterogamety.

113. Answer (1)

Hint: In case of X-linked recessive disorders, daughters must have affected father.

Sol.: Since individual I-2 and II-3 are affected and have different genotypes, the disorder can only be a dominant disorder. The probability of III-1 being homozygous for myotonic dystrophy is zero because she is affected but has an unaffected father.

114. Answer (2)

Hint: The number of phosphodiester linkages is equal to the number of nitrogenous bases present in circular DNA.

Sol.: The genome of $\phi \times 174$ phase is a single stranded circular DNA consisting of 5386 nucleotides.

115. Answer (3)

Hint: The anticodon can form hydrogen bonds with the codon.

Sol.: The 3′ end of tRNA is amino acid acceptor end.

116. Answer (2)

Hint: Severo Ochoa enzyme is also called polynucleotide phosphorylase.

Sol.: 23S rRNA is a catalytic RNA involved in peptide bond formation, also known as peptidyl transferase.

117. Answer (1)

Hint: 0.1% of human genome is unique to each human being.

Sol.:

Genes discovered post HGP that have yet unknown functions	–	More than 50 per cent
Genomic content that codes for protein	–	Less than 2 per cent
Nucleotide sequence identical across all humans	–	More than 99 per cent
Base sequence that may vary in the genomes of two people	–	Less than 0.1 per cent

118. Answer (3)

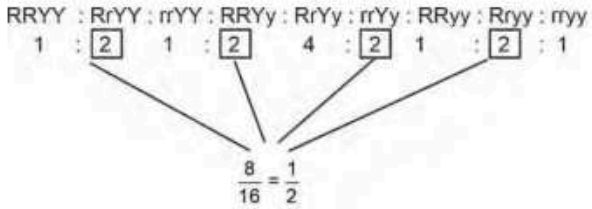
Hint: Independent segregation of two non-allelic genes is called independent assortment.

Sol.: Independent assortment only occurs when two genes are located on separate pairs of homologous chromosomes.

119. Answer (1)

Hint: The dihybrid genotypic ratio of F_2 generation is 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1.

Sol.: The F_2 genotypic ratio of a dihybrid cross can be written as following. The progeny that is homozygous for one and heterozygous for the other character are also shown.



120. Answer (3)

Hint: Formation of gamete and zygote can be understood by a Punnett square.

Sol.:

Discovery of X-body in insects	–	Henking
Graphical representation to show all possible genotypes in offspring	–	Reginald C. Punnett
Experimental verification of chromosomal theory of inheritance	–	T.H. Morgan
Mapped the position of genes on chromosome	–	Alfred Strutevant

121. Answer (4)

Hint: Phenotypic categories can be determined by the formula 2^n.

Sol.: The parent TtRr would produce 4 types of gametes leading to 4 different phenotypes in the progeny.

122. Answer (1)

Hint: Sequencing the whole set of genome and later assigning different regions in the sequence with functions is sequence annotation.

Sol.: Expressed Sequence Tags (ESTs) are focused on identifying all the genes that are expressed as RNA.

123. Answer (2)

Hint: *i*-gene forms mRNA leading to synthesis of repressor.

Sol.: Structural genes -3 (*lac* z, *lac* y, *lac a*)

mRNA-2 (from structural genes and *i* gene)

Enzymes-3 (β-galactosidase, permease and transacetylase)

124. Answer (1)

Hint: Nuclein refers to DNA.

Sol.: Nuclein (DNA) and Nucleoid (region of DNA in prokaryotes) can be present in *E. coli* together.

125. Answer (1)

Hint: A nucleosome consists of histones and DNA.

Sol.: Prokaryotes lack histones as well as nucleosomes.

126. Answer (1)

Hint: 16S rRNA is found in 30S subunit of ribosome.

Sol.: 16S rRNA is found in 70S ribosomes. It is not formed by RNA polymerase I.

127. Answer (3)

Hint: Down's syndrome occurs due to trisomy of chromosome 21.

Sol.:

(A) An individual with Turner's syndrome-1 X chromosome

(B) Female grasshopper-2 X chromosomes

(C) A female with Down's syndrome-2 X chromosomes

(D) An individual with Klinefelter's syndrome-2 X chromosomes

128. Answer (4)

Hint: There are total 64 genetic codons.

Sol.: Minimum number of codons required is one per amino acid. So for 20 amino acids, 20 codons are required.

129. Answer (2)

Hint: Out of 64 genetic codons, three does not code for any amino acids.

Sol.: The genetic codons UAA, UAG and UGA are stop codons. UAU codes for tyrosine.

130. Answer (3)

Hint: Sanger developed the technique to determine sequence of nucleotides in nucleic acid.

Sol.: Sanger developed the technique to determine sequence of amino acids in protein.

131. Answer (1)

Hint: Chromosomal theory of inheritance was proposed is 1902.

Sol.: Chromosomal theory of inheritance was proposed by Sutton.

132. Answer (3)

Hint: Law of dominance is not universal.

Sol.: The F_1 heterozygous tall progeny would form gametes T and t in equal proportion. *Antirrhinum* (Snapdragon) shows incomplete dominance for flower colour.

133. Answer (4)

Hint: Honeybee shows haplodiploid sex determination.

Sol.: Klinefelter's syndrome is a male specific disorder and has zero probability in females. Haemophilia in females is rare but not impossible.

Drone in honeybee are developed from unfertilised eggs.

Intensity of α-thalassemia increases depending on how many of the alleles are affected by deletion.

134. Answer (2)

Hint: Sahiwal cows are Indian breeds.

Sol.: Sahiwal cows were developed from wild cows via exploitation and artificial selection of certain desirable traits.

135. Answer (2)

Hint: Mendel started his research in 1856.

Sol.: Mendel published his research in the year 1865.

[ZOOLOGY]

136. Answer (2)

Hint: Cause for population explosion

Sol.: Counselling and creating awareness among people about reproductive organs, adolescence and associated changes, safe and hygienic sexual practices, STIs including AIDS, *etc.*, are the primary steps towards reproductive health. Providing medical facilities and care to the problems like menstrual irregularities, pregnancy related aspects, delivery, medical termination of pregnancy, birth control, infertility, post-natal child and maternal management are important aspects of the Reproductive and Child Health Care programmes.

137. Answer (4)

Hint: Exclude the viral diseases

Trichomoniasis	–	Protozoal STI caused by *Trichomonas vaginalis*
AIDS	–	Viral STI caused by Human Immunodeficiency Virus
Hepatitis–B	–	Viral STI caused by Hepatitis-B Virus
Chlamydiasis	–	Bacterial STI caused by *Chlamydia trachomatis*

138. Answer (3)

Hint: Identify a copper releasing IUD

Sol.: Progestasert and LNG-20 are hormone releasing IUDs that make the uterus unsuitable for implantation and the cervix hostile to the sperms. Progestogens alone or in combination with estrogen can be used by females as injections or implants under the skin.

Multiload-375 is a copper releasing IUD.

139. Answer (2)

Hint: White winged moths did not survive.

Sol.: During post-industrialisation period, the tree trunks became dark due to industrial smoke and soots. Under this condition, the white-winged moth did not survive due to predators, dark-winged or melanised moth survived. Before industrialisation set in, thick growth of almost white-coloured lichen covered the trees-in that background the white winged moths survived but the dark-coloured moth were picked out by predators.

140. Answer (2)

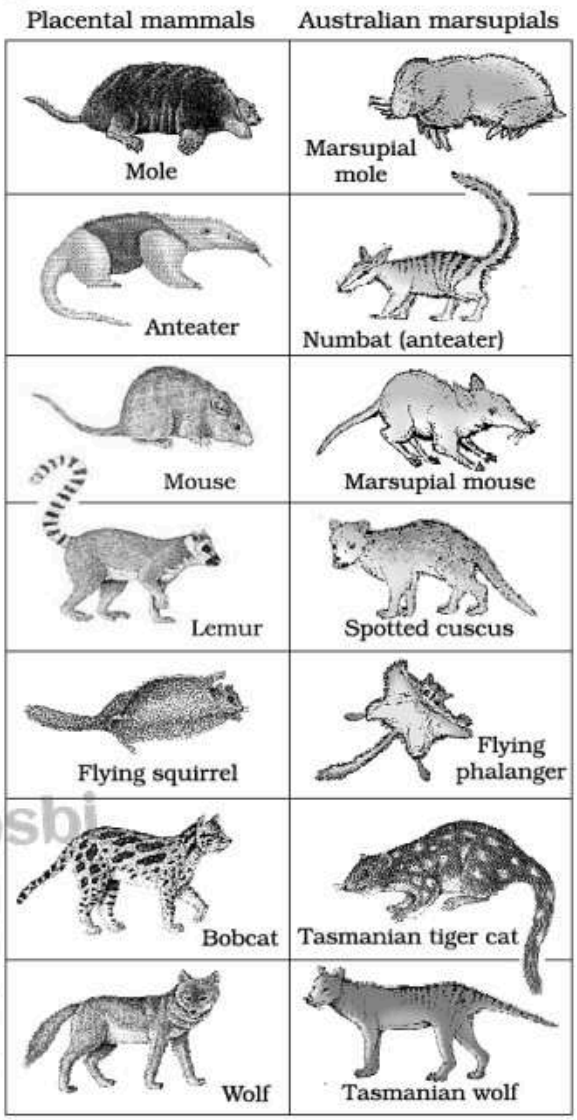
Hint: Eliminate the concept given by Hugo de Vries.

Sol.: A French naturalist Lamarck had said that evolution of life forms had occurred but driven by use and disuse of organs. He gave the example of Giraffes who in an attempt to forage leaves on tall trees had to adapt by elongation of their necks. As they passed on this acquired character of elongated neck to succeeding generations, Giraffes, slowly, over the years, came to acquire long necks. Nobody believes this conjecture any more.

141. Answer (1)

Hint: Include the organism that shows convergent evolution with flying phalanger.

Sol.:



142. Answer (3)

Hint: True for condoms

Sol.: In barrier methods, ovum and sperms are prevented from physically meeting with the help of barriers. Condoms are barriers made of thin latex/rubber sheath that are used to cover the penis in the male or vagina and cervix in the female reproductive tract. Use of condoms has increased in recent years due to its additional benefit of protecting the user from contracting STIs.

Cu7 is a copper releasing IUD.

In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse the fetal cells.

143. Answer (1)

Hint: Same mode of action exhibited by implants.

Sol. Administration of progestogens or progestogen-estrogen combination or IUD within 72 hours of coitus have been found to be very effective as emergency contraceptives as they could be used to avoid possible pregnancy due to rape or casual unprotected intercourse.

144. Answer (3)

Hint: Ovulation takes place on 14th day.

Sol. Periodic abstinence is an example of natural method of contraception. It is the method in which the couples avoid or abstain from coitus from day 10 to 17 of the 28 days menstrual cycle when ovulation could be expected. As chances of fertilisation are very high during this period, it is called the fertile period. Thus, by abstaining from coitus during this period, conception could be prevented.

145. Answer (4)

Hint: False for viral STIs

Sol. Infections or diseases which are transmitted *via* sexual intercourse are collectively called sexually transmitted infections (STIs) or venereal diseases. Except for hepatitis-B, genital herpes and HIV infections other STIs are completely curable if detected early and treated properly.

146. Answer (2)

Hint: Opinion of one medical practitioner is required before 12th week.

Sol. According to the MTP (Amendment) Act, 2017, a pregnancy may be terminated on certain grounds within the first 12 weeks of pregnancy on the opinion of one registered medical practitioner. If the pregnancy has lasted more than 12 weeks, but fewer than 24 weeks, two registered medical practitioners must be of the opinion, formed in good faith, that the required ground exist.

147. Answer (2)

Hint: Natural methods of contraception

Sol. Natural methods work on the principle of avoiding chances of ovum and sperms meeting. They do not involve any external devices, chemical, or hormones. IUDs such as Multiload 375, LNG-20 primarily increase sperm phagocytosis or release ions/hormones to prevent implantation. Barrier methods prevent the meeting of gametes; they are mechanical devices (made of rubber/latex) and often used with spermicidal chemicals, unlike natural methods.

Hormonal methods work by inhibiting ovulation and altering the uterine lining.

148. Answer (3)

Hint: Primitive atmosphere was reducing.

Sol. In 1953, S.L. Miller, an American scientist created conditions in a laboratory scale. He created electric discharge in a closed flask containing CH_4, H_2, NH_3 and water vapour at 800°C. He observed formation of amino acids. In similar experiments other observed, formation of sugars, nitrogenous bases, pigment and fats.

149. Answer (1)

Hint: Heterozygote frequency = 2pq

Sol.: If 320 individuals out of 2000 show the recessive phenotype:

$q^2 = 320/2000 = 0.16$

$q = 0.4$

According to the Hardy Weinberg equation:

$p + q = 1 \Rightarrow p = 0.6$

Heterozygote frequency = $2pq = 2 \times 0.6 \times 0.4$

$= 0.48$

150. Answer (3)

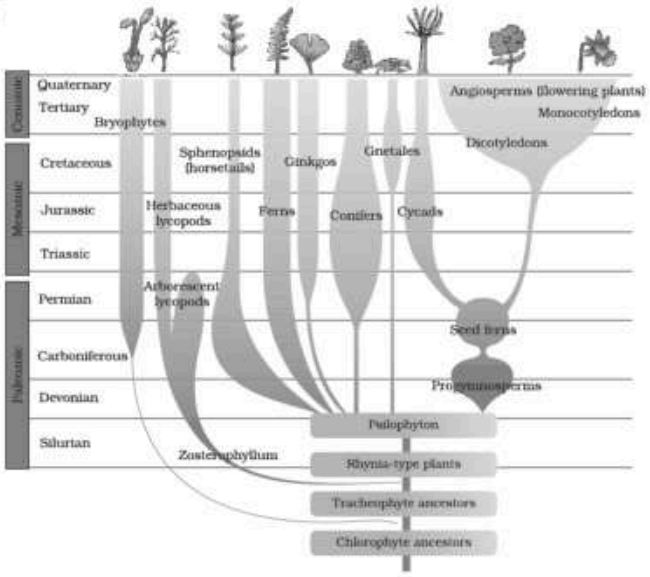
Hint: Biggest dinosaur

Sol.: About 65 mya, the dinosaurs suddenly disappeared from the Earth. *Tyrannosaurus rex* was about 20 feet in height and had huge fearsome dagger-like teeth. The first non-cellular forms of life could have originated 3 billion years back.

151. Answer (2)

Hint: Progymnosperms and gnetales share the same ancestor.

Sol.:



152. Answer (4)

Hint: Example of convergent evolution

Sol.: The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called adaptive radiation. Darwin's finches represent one of the best example of this phenomenon. Another example is Australian marsupials.

When more than one adaptive radiation appeared to have occurred in an isolated geographical area (representing different habitats), one can call this convergent evolution.

153. Answer (3)

Hint: Homology is the result of divergent evolution

Sol.: Homology is based on divergent evolution whereas, analogy refers to a situation exactly opposite. Wings of butterfly and of birds look alike. They are not anatomically similar structures though they perform similar functions. Hence, analogous structures are the result of convergent evolution.

154. Answer (3)

Hint: Different structures evolving for the same function and hence having similarity.

Sol.: Analogous structures are the result of convergent evolution-different structures evolving for the same function and hence having similarity. Other examples of analogy are the eye of the *Octopus* and of mammals. It is the similar habitat that has resulted in selection of similar adaptive features in different groups of organisms but towards the same function.

155. Answer (3)

Hint: Biogenetic law suggests that ontogeny repeats phylogeny.

Sol.: Embryological support for evolution was proposed by Ernst Heckel based upon the observation of certain features during embryonic stage common to all vertebrates that are absent in adult. For example, the embryos of all vertebrates including human develop a row of vestigial gill slit just behind the head but it is a functional organ only in fish and not found in any other adult vertebrates. However, this proposal was disapproved on careful study performed by Karl Ernst von Baer. He noted that embryos never pass through the adult stages of other animals.

156. Answer (2)

Hint: Exclude the techniques that involve *in-vitro* fertilisation.

Sol.: **ZIFT (Zygote Intra-Fallopian Transfer):** Involves *in-vitro* fertilisation where the zygote is later transferred into the fallopian tube.

IUT (Intra-Uterine Transfer): Involves transferring of an embryo (formed *in-vitro*) into the uterus.

ICSI (Intra-Cytoplasmic Sperm Injection): A specialized *in-vitro* procedure where a single sperm is directly injected into an ovum in a laboratory.

157. Answer (1)

Hint: Cranial capacity of neanderthal man was 1400 cc.

Sol. Cranial capacity of *Homo habilis* – 650 to 800 cc.

Cranial capacity of *Australopithecus* – 500 cc

Cranial capacity of *Homo erectus* – 900 cc

158. Answer (3)

Hint: Happened around 2000 mya

Sol. The first cellular forms of life appeared about 2000 million years ago. Fossils of *Homo erectus* were discovered in Java in 1891. Sea weeds and a few primitive plants existed probably around 320 mya.

159. Answer (3)

Hint: Copper releasing IUD

Sol.: Saheli is a non-steroidal oral contraceptive pill developed by CDRI, Lucknow. It is a "once a week" pill with very few side effects. Cu7 is a copper releasing IUD, not hormone releasing IUD. It works by releasing copper ions that suppresses sperm motility and their fertilizing capacity Hormone releasing IUDs include Progestasert and LNG-20. Vasectomy and tubectomy are sterilization or terminal methods. They are highly effective but their reversibility is very poor.

160. Answer (4)

Hint: True for implants

Sol.: Reproductive Tract Infections (RTIs) specifically affect reproductive organs. However, some sexually transmitted infections (STIs) like AIDS and Hepatitis-B primarily affect the immune system or liver. MTP is considered safe during the first trimester. The risk to the mother increases significantly during the second trimester. IUDs inhibit implantation but not ovulation and insemination.

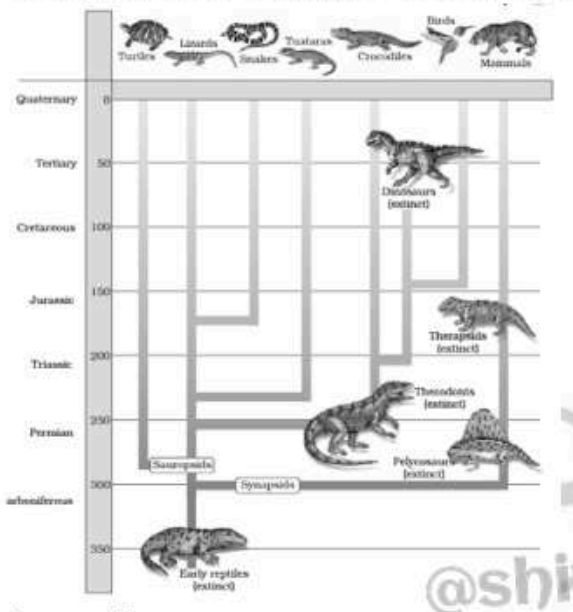
161. Answer (2)

Hint: Exclude assortative mating.

Sol. Five factors are known to affect Hardy Weinberg equilibrium. These are gene migration or gene flow, genetic drift, mutation, genetic recombination and natural selection.

162. Answer (2)

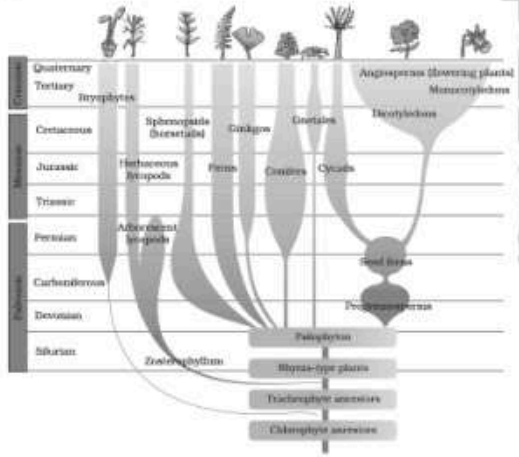
Hint: Exclude the one that gave rise to dinosaurs



163. Answer (2)

Hint: Exclude the descendants of psilophyton.

Sol.:



164. Answer (2)

Hint: Decrease in MMR indicates improved maternal health services.

Sol.: A rapid decline in death rate, maternal mortality rate (MMR) and infant mortality rate (IMR) as well as an increase in number of people in reproducible age are probable reasons for population explosion.

165. Answer (1)

Hint: Basis of lactational amenorrhea

Sol.: Breast feeding stimulates prolactin secretion and evidence suggests that prolactin inhibits GnRH secretion and thus inhibits the secretions of gonadotropins.

166. Answer (3)

Hint: The Neanderthal man buried their dead.

Sol.: During ice age, between 75,000-10,000 years ago modern *Homo sapiens* arose. Pre-historic cave art developed about, 18,000 years ago. One such cave paintings by pre-historic humans can be seen at Bhimbetka rock shelter in Raisen district of Madhya Pradesh. Agriculture came around 10,000 years back and human settlements started.

167. Answer (2)

Hint.: Exclude the one who gave the theory of inheritance of acquired character.

Sol.: It is possible that the work of Thomas Malthus on populations influenced Darwin. Natural selection is based on certain observations which are factual. Lamarck proposed the " theory of use and disuse of organs", which is a different evolutionary mechanism. Hugo de Vries is associated with the mutation theory, suggesting evolution occurs through sudden, large changes (saltation).

168. Answer (3)

Hint: Exclude the theory which suggests that Earth is about 4000 years old.

Sol.: Louis Pasteur by careful experimentation demonstrated that life comes only from pre-existing life. He showed that in pre-sterilised flasks, life did not come from killed yeast while in another flask open to air, new living organisms arose from killed yeast. Spontaneous generation theory was dismissed once and for all. However, this did not answer how the first life form came on Earth.

169. Answer (3)

Hint: True for reptiles and birds

Sol.: The amphibians evolved into reptiles. They lay the shelled eggs which do not dry up in sun unlike those of amphibians.

170. Answer (2)

Hint: Included under test tube baby programme

Sol.: In test tube baby programme, ova from the wife/donor (female) and sperms from the husband/donor (male) are collected and are induced to form zygote under simulated conditions

in the laboratory. The zygote or early embryos (with upto 8 blastomeres) could then be transferred into the fallopian tube (ZIFT-Zygote Intra Fallopian Transfer) and embryos with more than 8 blastomeres, into the uterus (IUT-Intra uterine transfer), to complete its further development.

171. Answer (3)

Hint: Venereal disease is just another term for STI.

Sol.: One could be free of STI by following the simple principles given below (i) Avoid sex with unknown partners/multiple partners. (ii) Always try to use condoms during coitus. (iii) In case of doubt, one should go to a qualified doctor for early detection and get complete treatment, if diagnosed with infection. Trichomoniasis is caused by a protozoan, *Trichomonas vaginalis*.

172. Answer (1)

Hint: Exclude the event that took place in 1951.

Sol. Government of India legalised MTP in 1971 with some strict conditions to avoid its misuse. Our population which was approximately 350 million at the time of our independence reached close to the billion marks by 2000 and crossed 1.2 billion in May 2011. India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal. These programmes called 'family planning' were initiated in 1951 and were periodically assessed over the past decades.

173. Answer (2)

Hint: Prevalent during Cenozoic era.

Sol.: Conifers were descendants of Psilophyton. Lichens can be used as industrial pollution indicators.

174. Answer (3)

Hint: Lived in East African grasslands.

Sol.: *Ramapithecus* was more man-like while *Dryopithecus* was more ape-like. Two mya, *Australopithecus* probably lived in East African grasslands. Evidence shows they hunted with stone weapons but essentially ate fruits. The Neanderthal man with a brain size of 1400 cc lived in near east and central Asia between 1,00,000-40,000 years back.

175. Answer (4)

Hint: A type of IUD.

Sol.: Diaphragms, cervical caps and vaults are barriers made of rubber that are inserted into the female reproductive tract to cover the cervix during coitus. Condoms are barriers made of thin rubber/latex sheath that are used to cover the penis in the male or vagina and cervix in the female.

176. Answer (1)

Hint: Vasectomy

Sol.: Surgical methods, also called sterilisation, are generally advised for the male/female partner as a terminal method to prevent any more pregnancies. Sterilisation procedure in the male is called 'vasectomy'. In vasectomy, a small part of the vas deferens is removed or tied up through a small incision on the scrotum.

177. Answer (3)

Hint: Results due to branching.

Sol.: Among the stories of evolution of individual species, the story of evolution of modern man most interesting and appears to parallel evolution of human brain and language.

Human evolution is viewed as a very slow, gradual process occurring over millions of years, leading eventually to a single surviving species (*Homo sapiens*).

178. Answer (4)

Hint: Ramification of genetic drift.

Sol.: The scenario described, where a population is drastically and randomly reduced by a natural calamity (volcanic eruption), is a classic example of the bottleneck effect. The bottleneck effect is a form of genetic drift where chance events cause a severe reduction in population size, and the surviving individuals have a gene pool that may not be representative of the original population's diversity. This leads to a significant loss of genetic variation.

179. Answer (2)

Hint: CuT is an example of copper releasing IUD.

Sol.: IUDs do not inhibit oogenesis and ovulation. Contraceptives are not regular requirement for the maintenance of reproductive health.

180. Answer (4)

Hint: Related to natural selection.

Sol.: According to Charles Darwin, the organic evolution is due to descent with modification of organisms.

❑ ❑ ❑

All India Aakash Test Series for NEET - 2027

TEST – 2 (Code-B)

Click here for Code-A Sol.

Test Date: 04/10/2026

ANSWERS

1.	(4)	37.	(2)	73.	(3)	109.	(1)	145.	(3)
2.	(1)	38.	(3)	74.	(1)	110.	(2)	146.	(2)
3.	(4)	39.	(1)	75.	(1)	111.	(3)	147.	(3)
4.	(3)	40.	(1)	76.	(1)	112.	(2)	148.	(3)
5.	(2)	41.	(1)	77.	(2)	113.	(1)	149.	(2)
6.	(2)	42.	(1)	78.	(1)	114.	(1)	150.	(3)
7.	(1)	43.	(3)	79.	(4)	115.	(3)	151.	(1)
8.	(2)	44.	(1)	80.	(1)	116.	(1)	152.	(2)
9.	(2)	45.	(4)	81.	(1)	117.	(1)	153.	(2)
10.	(3)	46.	(3)	82.	(1)	118.	(2)	154.	(2)
11.	(4)	47.	(2)	83.	(3)	119.	(1)	155.	(2)
12.	(3)	48.	(1)	84.	(1)	120.	(1)	156.	(4)
13.	(4)	49.	(2)	85.	(2)	121.	(1)	157.	(3)
14.	(2)	50.	(2)	86.	(2)	122.	(3)	158.	(3)
15.	(4)	51.	(3)	87.	(4)	123.	(1)	159.	(1)
16.	(4)	52.	(1)	88.	(3)	124.	(2)	160.	(2)
17.	(2)	53.	(4)	89.	(4)	125.	(1)	161.	(3)
18.	(4)	54.	(2)	90.	(4)	126.	(2)	162.	(3)
19.	(4)	55.	(3)	91.	(2)	127.	(1)	163.	(3)
20.	(3)	56.	(3)	92.	(2)	128.	(3)	164.	(4)
21.	(4)	57.	(4)	93.	(4)	129.	(1)	165.	(2)
22.	(3)	58.	(1)	94.	(3)	130.	(1)	166.	(3)
23.	(2)	59.	(4)	95.	(1)	131.	(3)	167.	(1)
24.	(4)	60.	(2)	96.	(3)	132.	(1)	168.	(3)
25.	(1)	61.	(2)	97.	(2)	133.	(4)	169.	(2)
26.	(4)	62.	(2)	98.	(4)	134.	(4)	170.	(2)
27.	(2)	63.	(4)	99.	(3)	135.	(3)	171.	(4)
28.	(3)	64.	(4)	100.	(1)	136.	(4)	172.	(3)
29.	(1)	65.	(2)	101.	(1)	137.	(2)	173.	(1)
30.	(3)	66.	(4)	102.	(1)	138.	(4)	174.	(3)
31.	(2)	67.	(2)	103.	(2)	139.	(3)	175.	(1)
32.	(4)	68.	(1)	104.	(1)	140.	(1)	176.	(2)
33.	(4)	69.	(3)	105.	(4)	141.	(4)	177.	(2)
34.	(2)	70.	(1)	106.	(3)	142.	(3)	178.	(3)
35.	(2)	71.	(4)	107.	(1)	143.	(2)	179.	(4)
36.	(1)	72.	(4)	108.	(3)	144.	(1)	180.	(2)

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HINTS & SOLUTIONS

[PHYSICS]

1. Answer (4)

Hint: In uniform magnetic field, when particle is thrown at an angle with magnetic field, then path of particle is helical with uniform pitch and

Pitch = $v_{||}T$ and $R = \frac{mv_{\perp}}{qB}$

Sol.: $v_{||} = v\cos\theta$ & $v_{\perp} = v\sin\theta$

$$a = \frac{\cos 37^\circ}{\cos 53^\circ} = \frac{4\times 5}{5\times 3} = \frac{4}{3}$$

$$b = \frac{\sin 37^\circ}{\sin 53^\circ} = \frac{3}{5}\times\frac{5}{4} = \frac{3}{4}$$

$$\therefore \quad a+b = \frac{4}{3}+\frac{3}{4} = \frac{16+9}{12} = \frac{25}{12}$$

$$a-b = \frac{4}{3}-\frac{3}{4} = \frac{16-9}{12} = \frac{7}{12}$$

$$a\times b = \frac{4}{3}\times\frac{3}{4} = 1$$

2. Answer (1)

Hint: For net force to be zero in both electric and magnetic field, $\vec{E} \,||\, \left(\vec{B}\times\vec{V}\right)$

Sol.: $\vec{E} \,||\, \left(\vec{B}\times\vec{V}\right)$

$\therefore \quad \hat{i} \,||\, \left(\vec{B}\times\left(\hat{j}\right)\right.$

$\therefore$ Direction of magnetic field should be along + z axis

3. Answer (4)

Hint & Sol.: Ratio of magnitude of magnetic moment to that of its angular momentum of orbiting charge is $\frac{q}{2m}$.

4. Answer (3)

Hint: Net force on particle is given by, $\vec{F}_{net} = q\left(\vec{v}\times\vec{B}\right)+q\vec{E}$

Sol.: Since particle is initially at rest, it will accelerate in the direction of electric field which is parallel to magnetic field. Therefore, magnetic force on particle is zero. The locus of particle will be straight line.

5. Answer (2)

Hint: V_{ab} = Potential difference across 5 V battery
$= 5 - i$

Sol.: $5 - i = 0 \Rightarrow i = 5$ A

$$\Rightarrow \quad i = \frac{25}{3+R} = 5$$

$\Rightarrow \quad 5 = 3 + R$

$R = 2\ \Omega$

6. Answer (2)

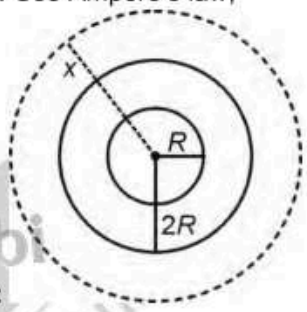
Hint: Brightness is proportional to power across lamp.

Sol.: When switch is closed, R_{eq} decrease
$\Rightarrow$ Current increases

$\therefore$ Power across A increases & across B decreases.

7. Answer (1)

Hint: Use Ampere's law,



Sol.:

$$i_{enc} = \int JdA$$

$$= \int_R^{2R} J_0 r^2 .2\pi r dr$$

$$= J_0 2\pi \int_R^{2R} r^3 dr$$

$$= J_0 2\pi \left[\frac{r^4}{4}\right]_R^{2R}$$

$$= \frac{J_0\pi}{2}\left[16R^4 - R^4\right].$$

$$i_{enc} = \frac{J_0\pi\times 15R^4}{2}$$

Now,

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 i_{enc}$$

$$B2\pi x = \frac{\mu_0\times 15}{2} J_0\pi R^4$$

$$B = \frac{15}{4x}\mu_0 J_0 R^4$$

8. Answer (2)

Hint: Use Ampere's law

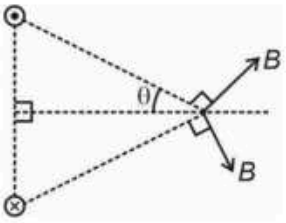
Sol.: Magnetic field due to hollow metallic pipe, $B = 0$ for $r < R$

$B = \frac{\mu_0 i}{2\pi r}$ for $r > R$

9. Answer (2)

Hint: Magnetic field due to long wire at a distance r is given by, $B = \frac{\mu_0 i}{2\pi r}$

Sol.:



$\vec{B}_{net} = 2B\sin\theta$

$= \frac{2\times\mu_0 i}{2\pi(4)}\times\frac{1}{4}$

$= \frac{4\pi\times10^{-7}\times2}{16\pi}$

$= 5 \times 10^{-8}$ T $= 0.05\ \mu$T

10. Answer (3)

Hint: This system behaves like solenoid and magnetic field inside long solenoid is given by $\mu_0 ni$

Sol.: $B = \mu_0 ni$

$= 4\pi\times10^{-7}\times\frac{100}{2\pi\times10\times10^{-3}}\times2$

$= 4\times10^{-3}$ T $= 4$ mT

11. Answer (4)

Hint: Use concept of Ampere's law

Sol.: In Ampere's law, $\oint\vec{B}\cdot d\vec{l} = \mu_0 i_{enc}$, B is due to all the current present inside and outside the loop.

12. Answer (3)

Hint: Magnetic field due to semi infinite wire, is $\frac{\mu_0 i}{4\pi d}$ and it is zero on its own axis

Sol.: $\vec{B}_p = \frac{\mu_0 i}{4\pi r}\otimes$

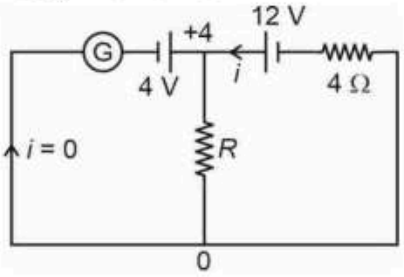
13. Answer (4)

Hint & Sol.: Potential difference across a battery is more than its emf when battery is charging

14. Answer (2)

Hint: Use, Kirrchhoff's law.

Sol.:



$iR = 4$ ...(1)

$\Rightarrow 12 - 4i = 4$

$\Rightarrow 4i = 8$

$\Rightarrow i = 2$ A

$\Rightarrow$ Put value in equation (1)

$R = 2\ \Omega$

15. Answer (4)

Hint and Sol.: Bismuth (Bi), copper (Cu) and Salt (NaCl) are diamagnetic substances.

Copper chloride ($CuCl_2$) is paramagnetic substance.

16. Answer (4)

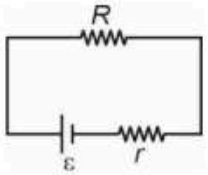
Hint: As i changes in same resistor, charge density (n) remains constant.

Sol.: $i = JA \Rightarrow J$ becomes double

$i = neAvd \Rightarrow V_d$ becomes double

17. Answer (2)

Hint: When cell is connected through external resistance R, as shown, then



$i = \frac{\varepsilon}{R+r}$

Sol.: $i = \frac{\varepsilon}{R+r}$

$1 = \frac{\varepsilon}{4+r} \Rightarrow \varepsilon = 4 + r$ ...(i)

Now, $\frac{1}{2} = \frac{\varepsilon}{10+r}$

$\Rightarrow \frac{10+r}{2} = 4 + r$ [From equation (i)]

$\Rightarrow 10 + r = 8 + 2r$

$\Rightarrow r = 2\ \Omega$

18. Answer (4)

Hint: For paramagnetic material, $I \propto \frac{H}{T}$

Sol.: $B = \mu H$

$$\frac{I_2}{I_1} = \frac{B_2}{B_1} \times \frac{T_1}{T_2} = \frac{1}{2} \times \frac{10}{20} = \frac{1}{4}$$

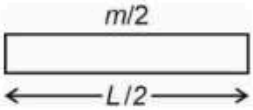
$$I_2 = \frac{I_1}{4} = \frac{0.6}{4} = 0.15 \text{ A/m}$$

19. Answer (4)

Hint: Time period of oscillation of needle in magnetic field is given by, $T = 2\pi\sqrt{\frac{I}{MB}}$

Sol.: Initially,  $I = \frac{mL^2}{12}$

Now



$$I' = \frac{\frac{m}{2}\left(\frac{L}{2}\right)^2}{12} = \frac{m}{2} \times \frac{L^2}{4 \times 12} = \frac{I}{8}$$

$$M' = \frac{M}{2}$$

$$T' = 2\pi\sqrt{\frac{I}{8 \times \frac{M}{2} \times B}} = \frac{T}{2}$$

$T' = 2$ s

20. Answer (3)

Hint: Susceptibility, $\chi = \mu_r - 1$ and $B = \mu_r\mu_0 H$

Sol.: $\mu_r = \chi + 1$

$= 550$

$\phi = BA = \mu_r\mu_0 HA$

$= 550 \times 4\pi \times 10^{-7} \times 10^3 \times 2 \times 10^{-5}$

$= 44\pi \times 10^{-7}$ Wb $= 4.4\pi \times 10^{-6}$ Wb

21. Answer (4)

Hint & Sol.: Since $B > B_0$, therefore material can be paramagnetic or ferromagnetic because intensity of magnetising field is positive.

22. Answer (3)

Hint: Energy stored across capacitor, $U = \frac{1}{2}CV^2$

Sol.: $V = 100$ V

$$U = \frac{1}{2}CV^2 = \frac{1}{2} \times 10 \times 10^{-6} \times 10^4$$

$= 5 \times 10^{-2} = 50$ mJ

23. Answer (2)

Hint: Charge transferred, $q = \int i\,dt$

Sol.: $q = \int_0^2 (2 + 4t)\,dt$

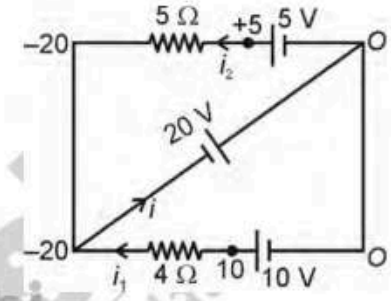
$= 2(t)_0^2 + 2\left(t^2\right)_0^2$

$= 4 + 8 = 12$ C

24. Answer (4)

Hint: Use Kirchhoff's law

Sol.:



$$i_1 = \frac{30}{4} = 7.5 \text{ A}$$

$$i_2 = \frac{25}{5} = 5 \text{ A}$$

$i = i_1 + i_2$

$= 7.5 + 5$

$= 12.5$ A

25. Answer (1)

Hint: $\varepsilon_{ind} = \frac{-d\phi}{dt}$

Sol.: At $0 < t < \frac{l}{v}$

Flux is increasing downwards, so emf induced will be anticlockwise

At $\frac{l}{v} < t < \frac{3l}{v}$

Flux is constant

So, induced emf is zero.

At $\frac{3l}{v} < t < \frac{4l}{v}$

Flux is decreasing downwards, so emf induced will be clockwise.

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At $t > \frac{4l}{v}$

Flux is zero, so no emf will be induced.

Also, 'v' is uniform, so induced emf in the wire will be constant.

26. Answer (4)

Hint & Sol.:

a. The intensity of magnetisation (I) is very small, negative and is directly proportional to magnetising field (H) for diamagnetic substance.

b. In non-uniform magnetic field, the paramagnetic substances are attracted towards the stronger parts of magnetic field.

c. Substances, which at room temperature, retain their ferromagnetic property for a long period of time are called permanent magnets.

27. Answer (2)

Hint: At null point, it will behave as balanced wheatstone bridge

Sol.: $\frac{7}{25} = \frac{R}{75}$

$\Rightarrow R = 21\ \Omega$

28. Answer (3)

Hint & Sol.: Resistance, $R = \frac{\rho l}{A}$

It depends on dimensions of resistor.

29. Answer (1)

Hint & Sol.: For current through internal resistance to be maximum, equivalent resistance should be minimum, *i.e.*, $x = 0$.

30. Answer (3)

Hint: Only magnetic force $F_m = qV_0B_0$ acts as a centripetal force

Sol.: $F_m = qV_0B_0$

$$= qV_0\left(\frac{mV_0}{ql_0}\right) = \frac{mv_0^2}{l_0}$$

$\therefore$ Spring remains in its natural length

$\therefore$ Potential energy remains constant

$\Rightarrow$ Since only F_m is acting $\Rightarrow$ Kinetic energy remains constant

$\Rightarrow$ F_m acts as a central force or pass through O

Therefore, Angular momentum about point O remains conserved.

31. Answer (2)

Hint: $\vec{F}_{net} = q\vec{E} + q\left(\vec{V} \times \vec{B}\right)$

Sol.: $\vec{F}_E = q\vec{E} = 2\times10^{-9}\times2 = \left(4\times10^{-9}\hat{k}\right)$ N

$$\vec{F}_B = q(v\cos60°)B\,\hat{k} = 2\times10^{-9}\times2\times\frac{1}{2}\times2$$

$$= \left(4\times10^{-9}\hat{k}\right)\text{N}.$$

$$\vec{a} = \frac{\vec{F}_{net}}{m} = \frac{8\times10^{-9}\hat{k}}{10^{-6}} = \left(8\times10^{-3}\hat{k}\right)\ \text{ms}^{-2}$$

$$= 0.008\ \text{ms}^{-2}\ \hat{k}$$

32. Answer (4)

Hint & Sol.: For diamagnetic material, χ_m is independent of temperature.

33. Answer (4)

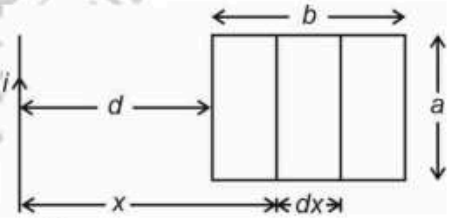
Hint: Magnetic field at a distance r at perpendicular bisector is, $B = \frac{-\mu_0\vec{M}}{4\pi r^3}$

Sol.: $B_{net} = 2\left(\frac{\mu_0 M}{4\pi\left(\frac{a}{\sqrt{2}}\right)^3}\right) = \frac{4\sqrt{2}\mu_0 M}{4\pi a^3} = \frac{\sqrt{2}\mu_0 M}{\pi a^3}$

34. Answer (2)

Hint: Mutual inductance, $M = \frac{\phi_2}{i_1}$

Sol.:



$$\int d\phi_2 = \int_d^{d+b}\frac{\mu_0 i}{2\pi x}a\,dx \Rightarrow \phi_2 = \frac{\mu_0 ia}{2\pi}\left[\ln(x)\right]_d^{d+b}$$

$$= \frac{\mu_0 ia}{2\pi}\ln\left(\frac{d+b}{d}\right)$$

$\Rightarrow M = \frac{\phi_2}{i} = \frac{\mu_0(2)}{2\pi}\ln\left(\frac{5+3}{5}\right)$

$\Rightarrow M = \frac{\mu_0}{\pi}\ln\left(\frac{8}{5}\right)$

35. Answer (2)

Hint: For induced emf to be zero, flux should be constant.

Sol.: $BlVt$ = Constant

$B = \frac{C}{lvt}$

36. Answer (1)

Hint: Range of ammeter is given by

$$\Rightarrow i = i_g\left(1+\frac{R_G}{s}\right)$$

Sol.: $i = i_g\left(1+\frac{R_G}{s}\right)$

$10 = 20\times10^{-3}\left(1+\frac{25}{s}\right)$

$\Rightarrow \quad 500 = 1+\frac{25}{s}$

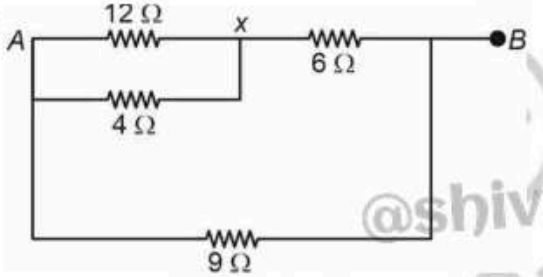
$\Rightarrow \quad \frac{25}{s} = 499$

$s = \frac{25}{499}\Omega$

37. Answer (2)

Hint: Rearrange the circuit by marking potential of points

Sol.:



$R_{AB} = \frac{9}{2}\ \Omega = 4.5\ \Omega$

38. Answer (3)

Hint: Total charge flown

$q = \frac{\Delta\phi}{R}$ & $\Delta\phi = \vec{B}\cdot\Delta\vec{A}$

Sol.: $\Delta\phi = qR = 10^{-5}\times 20 = 2\times10^{-4}$ Wb

&

$\Delta\phi = N\left(\vec{B}\cdot\Delta\vec{A}\right) = N\left(B\Delta A\right)$ [Angle b/w B & A is 0°]

$B = \frac{\Delta\phi}{N\Delta A} = \frac{2\times10^{-4}}{20\times2\times10^{-2}} = \frac{10^{-2}}{20} = 5\times10^{-4}$ T

39. Answer (1)

Hint: Due to change in current in straight wire, flux is changing in rectangular loop. Use Lenz's law to find induced current in loop.

Sol.:

At time = $t_1 \rightarrow \frac{di}{dt} = +\text{ve} \Rightarrow$ Direction of induced current in anticlockwise

At time = $t_2 \rightarrow \frac{di}{dt} = -\text{ve} \Rightarrow$ Direction is clockwise

At time = $t_3 \rightarrow \frac{di}{dt} = +\text{ve} \Rightarrow$ Direction is anticlockwise

40. Answer (1)

Hint: $F = mB$

Sol.: Unit of m is A m

$B = \frac{F}{m} = \frac{\text{N.m}}{\text{A m}^2} = \frac{\text{J}}{\text{A m}^2}$

41. Answer (1)

Hint: Resistance, $R = \frac{\rho l}{A}$

Sol.: $R = \frac{\rho l}{A} = \frac{\rho l^2}{V}$

$R \propto l^2 \ \Rightarrow \frac{100}{R'} = \frac{l^2}{4l^2} \ \Rightarrow R' = 400\ \Omega$

42. Answer (1)

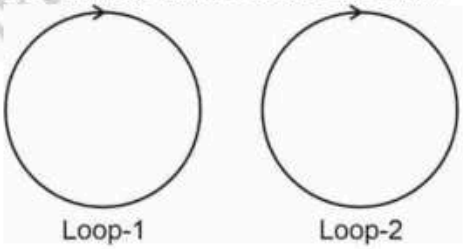
Hint: Magnitude of induced emf, $|\varepsilon| = \left|\frac{d\phi}{dt}\right|$

Sol.: $|\varepsilon| = \left|\frac{d\phi}{dt}\right| = 20$ V

43. Answer (3)

Hint: Use Lenz's law.

Sol.: Let current flows in loop-1 in clockwise direction. To oppose this change current I is induced in loop-2 in clockwise direction.



∴ Loop-2 is repelled by the loop-1

44. Answer (1)

Hint & Sol.: When magnet is either approaching or receding, it changes the flux through loop, according to Lenz's law it oppose its motion.

45. Answer (4)

Hint: Flux, $\phi = \vec{B}\cdot\vec{A}$

Sol.: $\vec{A} = (16\ \text{m}^2)\hat{k}$

$\phi = \vec{B}.\vec{A}$

$\phi = B_0\left(13\hat{i}-7\hat{j}+2\hat{k}\right).\left(16\hat{k}\right)$

$\phi = 32B_0$ Wb

[CHEMISTRY]

46. Answer (3)

Hint: The complex species which contain d electron(s) show colour due to d-d transition.

Sol.: $[Co(H_2O)_6]^{3+}$ is diamagnetic in nature and is an inner orbital complex.

$[Ti(H_2O)_6]^{3+}$ contain one d electron. It is violet in colour due to d-d transition.

47. Answer (2)

Hint: $[PtCl_2(en)_2]^{2+}$ shows geometrical isomerism

Sol.: trans-$[PtCl_2(en)_2]^{2+}$ has plane of symmetry and is optically inactive.

cis-$[PtCl_2(en)_2]^{2+}$ exist as a pair of enantiomers. Each enantiomer is chiral and optically active in nature.

48. Answer (1)

Hint: Spin only magnetic moment

$(\mu_s) = \sqrt{n(n+2)}$ BM

n = Number of unpaired electron(s)

Sol.:

Ion	Number of unpaired electron(s)	μ(BM)
Fe^{3+}	5	5.92
Co^{2+}	3	3.87
Cr^{2+}	4	4.90
Ti^{3+}	1	1.73

49. Answer (2)

Hint: More is the effective nuclear charge more will be the hydration of ions and more will be the energy released.

Sol.:

Ions		$\Delta_{hyd}H°(M^{2+})$(kJ mol^{-1})
V^{2+}	–	1895
Cr^{2+}	–	1925
Co^{2+}	–	2079
Ni^{2+}	–	2121

50. Answer (2)

Hint: Highest oxide of Mn is Mn_2O_7 while highest Mn fluoride is MnF_4.

Sol.: V_2O_5 is amphoteric in nature, hence it reacts with both acid and alkali.

CrO is basic in nature while Cr_2O_3 is amphoteric.

51. Answer (3)

Hint: $2KMnO_4 \xrightarrow{513\ K} K_2MnO_4 + MnO_2 + O_2$

Sol.: $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

52. Answer (1)

Hint and Sol.: Correct order of ligand field strength of the given species is

$C_2O_4^{2-} > F^- > S^{2-} > I^-$

53. Answer (4)

Hint: For first order reactions $kt = 2.303\log\frac{[R]_0}{[R]}$

Sol.: $k = \frac{2.303}{t}\log\frac{0.20}{0.0125}$

or, $k = \frac{2.303}{20\times60}\log16$

or, $k = \frac{2.303}{20\times60}\log(2)^4$

or, $k = \frac{2.303\times4}{20\times60}\log2$

or, $k = 2.3 \times 10^{-3}\ s^{-1}$

$t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{2.3\times10^{-3}}\ s = 301.3\ s$

54. Answer (2)

Hint: Oder is applicable to elementary as well as complex reactions.

Sol.: Order of a reaction is experimental quantity. It can be zero and even a fraction but molecularity cannot be zero or a non-integer.

55. Answer (3)

Hint: For zero order reaction

$\text{rate} = -\frac{1}{3}\frac{d[A]}{dt} = k[A]^0$

Sol.: $[A]_0 - [A] = 3kt$

or, $t = \frac{1}{3k}\{[A]_0 - [A]\}$

or, $t = \frac{1}{3\times2\times10^{-2}}\{2.4 - 0.48\}$

or, $t = \frac{1.92}{3\times2\times10^{-2}} = 32\ s$

56. Answer (3)

Hint: $\log\frac{k_2}{k_1}=\frac{E_a}{2.303R}\left[\frac{T_2-T_1}{T_1T_2}\right]$

Sol.: Rate of reaction α rate constant of reaction.

$$\log\frac{4}{1}=\frac{E_a}{2.303R}\left[\frac{320-300}{320\times300}\right]$$

$$\text{or, } E_a=\frac{2.303R\times320\times300}{20}\times\log4$$

$$\text{or, } E_a=\frac{2.303\times8.314\times320\times300}{20}\times2\log2$$

or, $E_a = 55.1\ kJ\ mol^{-1}$

57. Answer (4)

Hint: rate of reaction

$$=-\frac{d[A]}{dt}=-\frac{1}{3}\frac{d[B]}{dt}=\frac{1}{2}\frac{d[C]}{dt}$$

Sol.: $$-\frac{1}{3}\frac{d[B]}{dt}=\frac{1}{2}\frac{d[C]}{dt}$$

$$\text{or, } -\frac{d[B]}{dt}=\frac{3}{2}\frac{d[C]}{dt}$$

$$\text{or, } -\frac{d[B]}{dt}=\frac{3}{2}\times4\times10^{-3}\ molL^{-1}s^{-1}$$

$$-\frac{d[B]}{dt}=6\times10^{-3}\ molL^{-1}s^{-1}$$

58. Answer (1)

Hint: General electronic configuration of lanthanoids is: $[Xe]4f^{1\text{-}14}\ 5d^{0-1}6s^2$

Sol.:

Ions		Electronic configuration	Number of unpaired electrons
$Er^{2+}(Z=68)$	:	$4f^{12}$	2
$Tm^{2+}(Z=69)$	:	$4f^{13}$	1
$Nd^{2+}(Z=60)$	:	$4f^{4}$	4
$Dy^{2+}(Z=66)$	:	$4f^{10}$	4

59. Answer (4)

Hint: More is the ligand field strength more is the value of crystal field splitting energy; more will be the frequency of light absorbed and lesser is the value of wavelength of light absorbed.

Sol.: Ligand field strength order:

$en > NH_3 > H_2O > F^-$

Wavelength of light absorbed:

$$[CoF_6]^{3-} > [Co(H_2O)_6]^{3+} > [Co(NH_3)_6]^{3+} > [Co(en)_3]^{3+}$$

60. Answer (2)

Hint: Potassium dichromate is orange coloured compound

Sol.:

Species		Colour
$K_2Cr_2O_7$	–	Orange
Mn_2O_7	–	Green
MnO_4^-	–	Purple
CrO_4^{2-}	–	Yellow

61. Answer (2)

Hint: In acidic medium dichromate is reduced to Cr^{3+} and Fe^{2+} is oxidised to Fe^{3+}

Sol.:

$$Cr_2O_7^{2-}+14H^++6Fe^{2+}\rightarrow2Cr^{3+}+6Fe^{3+}+7H_2O$$

62. Answer (2)

Hint: $\ln k=\ln A-\frac{E_a}{RT}$

Sol.: $$\ln e^3=5-\frac{7.47\times10^3}{8.3\times T}$$

$$\text{or, } 3=5-\frac{0.9\times10^3}{T}$$

$$\text{or, } \frac{0.9\times10^3}{T}=2$$

$$\text{or, } T=\frac{0.9\times10^3}{2}=450\ K$$

63. Answer (4)

Hint: A Catalyst catalyses the forward as well as the backward reactions to the same extent so that the equilibrium state remains same but is reached faster.

Sol.: A catalyst catalyses the spontaneous reaction but does not catalyse non-spontaneous reactions.

64. Answer (4)

Hint: Rate = $[A_2]^x[B_2]^y$

overall order of reaction x + y

Sol.: $B_2 \rightleftharpoons B+B$

$$K=\frac{[B][B]}{[B_2]}=\frac{[B]^2}{[B_2]}$$

$$\text{or, } [B]=K^{\frac{1}{2}}[B_2]^{\frac{1}{2}}$$

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Rate of the reaction is determined by the slow step.

$$\text{Rate} = K[A_2][B] = k[A_2]K^{\frac{1}{2}}[B_2]^{\frac{1}{2}}$$

$$\text{Rate} = = kK^{\frac{1}{2}}[A_2][B_2]^{\frac{1}{2}}$$

$$\text{overall order of reaction} = 1+\frac{1}{2} = 1.5$$

65. Answer (2)

Hint: $k = \frac{2.303}{t_2 - t_1}\log\frac{r_1}{r_2}$ for a first order reaction

Sol.: $k = \frac{2.303}{60}\log\frac{0.08}{0.04}$

$k = 1.15 \times 10^{-2}\ s^{-1}$

66. Answer (4)

Hint: Inversion of cane sugar takes place in excess of water in acidic medium.

Sol.: $2NH_3(g) \xrightarrow[\text{Pt catalyst}]{1130K} N_2(g) + 3H_2(g)$

Rate = $k[NH_3]^0 = k$

- All natural and artificial radioactive decay of unstable nuclei take place by first order kinetics.
- $\underset{\text{cane sugar}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{\Delta} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$

  Rate = $k[C_{12}H_{22}O_{11}]$

$$2H_2O_2 \xrightarrow[\text{alkaline medium}]{I^-} 2H_2O + O_2$$

Rate = $k[H_2O_2][I^-]$

67. Answer (2)

Hint: As the atomic number increases, the effective nuclear charge of lanthanoid ion increases hence the basic strength of its hydroxide decreases.

Sol.: Correct order of atomic number of the lanthanoid ions: $Er^{3+} > Tb^{3+} > Eu^{3+} > Pm^{3+}$

correct order of basic strength

$Pm(OH)_3 > Eu(OH)_3 > Tb(OH)_3 > Er(OH)_3$

68. Answer (1)

Hint: $[MnBr_4]^{2-}$ is tetrahedral in shape and is a high spin complex.

Sol.: Tetrahedral complexes are high spin complexes hence d electrons in Mn^{2+} are unpaired, giving a spin only magnetic moment of 5 electrons.

69. Answer (3)

Hint: The ions which contain partially filled d orbitals show colour in aqueous medium due to d-d transition.

Sol.: d-d transition is not observed in metal ions containing completely filled d-orbitals or empty *d* subshell.

Aquated Zn^{2+}, Sc^{3+}, Ti^{4+} do not show colour in aqueous medium. The ions which do not contain *d* electrons, do not show colour in aqueous medium.

70. Answer (1)

Hint: For zero order reaction $t_{1/2} = \frac{[R]_0}{2k}$

Sol.: If $A \rightarrow P$ follows second order kinetics then,

$$\frac{-d[A]}{dt} = k[A]^2$$

$$\text{unit of } k = \frac{mol L^{-1}s^{-1}}{mol^2L^{-2}} = mol^{-1}Ls^{-1}$$

71. Answer (4)

Hint: $[CoF_6]^{3-}$ is a high spin complex

Sol.: Crystal field stabilization energy for

$$[CoF_6]^{3-} = \left[\frac{-2}{5}\times 4 + \frac{3}{5}\times 2\right]\Delta_0$$

$= [-1.6 + 1.2]\Delta_0$

$= -0.4\Delta_0$

72. Answer (4)

Hint: Metal present in cyanocobalamine is cobalt

Sol.: The pigment responsible for photosynthesis, chlorophyll, is a coordination compound of magnesium.

Haemoglobin, the red pigment of blood, which acts as oxygen carrier is a coordination compound of iron.

Vitamin B_{12}, cyanocobalamine, the antipernicious anaemia factor, is a coordination compound of cobalt.

Wilkinson catalyst, $[(Ph_3P)_3RhCl]$ is the coordination compound of rhodium and is used for the hydrogenation of alkenes.

73. Answer (3)

Hint: It is a first order reaction, as the unit of rate constant is s^{-1}.

$$kt = 2.303\log\frac{(p_A^\circ)}{(p_A^t)}$$

Sol.:

	$A(g)$	$\rightarrow 2B(g)$	$+ C(g)$
Initial pressure	320	0	0
Pressure after time t	$(320 - x)$	2x	x

$320 - x + 2x + x = 920$

$2x = 600$

$x = 300$

$$t = \frac{2.303}{k}\log\frac{320}{20}$$

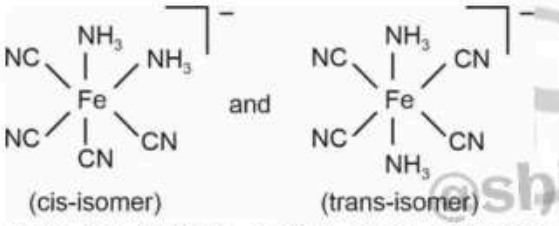
$$t = \frac{2.303}{2\times10^{-3}}\log16$$

$$t = \frac{2.303\times4\times0.3010}{2\times10^{-3}} = 1386 \text{ s}$$

74. Answer (1)

Hint: The species which contain plane of symmetry are optically inactive.

Sol.:



(cis-isomer) (trans-isomer)

Both the isomers contain plane of symmetry hence both are optically inactive.

75. Answer (1)

Hint: More is the effective nuclear charge, more is the ionisation enthalpy

Sol.:

Element	$\Delta_i H°$ (kJ mol^{-1})
Fe	762
Mn	717
Cr	653
V	650

76. Answer (1)

Hint: Most common oxidation state of vanadium is +5

Sol.: Most common oxidation state of nickel is +2

77. Answer (2)

Hint: Tetraammineaquachloridocobalt (III) chloride is $[Co(NH_3)_4(H_2O)Cl]Cl_2$

Sol.: 1 mole of $[Co(NH_3)_4(H_2O)Cl]Cl_2$ will give 2 moles of chloride ions in aqueous medium.

Mole of AgCl precipitated would be: $200 \times 0.4 \times 2 \times 10^{-3} = 0.16$

78. Answer (1)

Hint: More is the electron density on the metal better is the back donation from metal filled *d* orbital to the vacant π^* orbital of CO.

Sol.: Correct order of C–O bond strength of the given complexes is

$[Mn(CO)_6]^+ > [Ni(CO)_4] > [Cr(CO)_6] > [Co(CO)_4]^-$

79. Answer (4)

Hint: $E°$ value of Ce^{4+}/Ce^{3+} is 1.74 V. It is a strong oxidant.

Sol.: Lanthanoid metals on reaction with dilute acid liberate hydrogen gas.

80. Answer (1)

Hint: For stable electronic configuration, the metallic bonding is weak and the metals have a lower value of enthalpy of atomisation.

Sol.:

Metals	Enthalpy of atomisation (kJ mol^{-1})
V	515
Cr	397
Mn	281
Co	425

81. Answer (1)

Hint: Interstitial compounds are those which are formed when atoms like H, C or N are trapped inside the crystal lattices of metals.

Sol.: Interstitial compounds retain metallic conductivity.

82. Answer (1)

Hint: V_2O_5 catalyses the oxidation of SO_2 in the manufacture of H_2SO_4.

Sol.: Iron catalysts are used in the Haber process for the production of NH_3 from N_2/H_2 mixtures.

cis-platin and related compounds of platinum effectively inhibit the growth of tumours.

EDTA is used in the treatment of lead poisoning.

83. Answer (3)

Hint: Complexes in which a metal is bound to only one kind of donor groups are called homoleptic complexes.

Sol.: $K_3[Cr(C_2O_4)_3]$: potassium trioxalatochromate (III) is a homoleptic complex.

84. Answer (1)

Hint: $[MnCl_6]^{3-}$ is a high spin complex.

Sol.: $[MnCl_6]^{3-}$ is an octahedral high spin complex hence electronic configuration of Mn^{3+} in the complex will be $t_{2g}^3 e_g^1$.

85. Answer (2)

Hint: In $[Ni(CN)_4]^{2-}$, hybridization of Ni is dsp^2.

Sol.:

Complex	Hybridization	Shape	Magnetic nature
$[NiCl_4]^{2-}$	sp^3	Tetrahedral	Paramagnetic
$[Ni(CO)_4]$	sp^3	Tetrahedral	Diamagnetic
$[PtCl_4]^{2-}$	dsp^2	Square planar	Diamagnetic
$[Ni(CN)_4]^{2-}$	dsp^2	Square planar	Diamagnetic

86. Answer (2)

Hint: Ligand which can ligate through two different atoms is called ambidentate ligand.

Sol.: NO_2^- ion can coordinate either through nitrogen or through oxygen to a central atom/ion.

SCN^- ion can coordinate through either the sulphur or through nitrogen atom.

87. Answer (4)

Hint: Jahn Teller effect in the coordination complexes is observed when *d* orbitals are asymmetrically filled.

Sol.: In high spin d^4, d^6 and d^7 the electrons in the d orbitals are asymmetrically filled hence the complexes with these configurations show Jahn-Teller distortion.

88. Answer (3)

Hint: $\frac{K_{(T+10)}}{K_T} = (x)^{\Delta T/10}$

x = temperature coefficient

Sol.: $\frac{K_{(T+10)}}{K_T} = (2)^{50/10}$

$r_2 = r_1(2)^5$

89. Answer (4)

Hint: Rate = $k[A]^x \times [B]^y$

Sol.: $k[0.1]^x\ [0.2]^y = 1 \times 10^{-5}$ ...(i)

$k[0.2]^x\ [0.4]^y = 8 \times 10^{-5}$ ...(ii)

$k[0.2]^x\ [0.8]^y = 32 \times 10^{-5}$ ...(iii)

Dividing (i) by (ii)

$$\left(\frac{1}{2}\right)^x \times \left(\frac{2}{4}\right)^y = \frac{1}{8}$$

$$\left(\frac{1}{2}\right)^{x+y} = \frac{1}{8} = \left(\frac{1}{2}\right)^3$$

$x + y = 3$ ...(iv)

Dividing (ii) by (iii)

$$\left(\frac{4}{8}\right)^y = \frac{8}{32}$$

$$\left(\frac{1}{2}\right)^y = \left(\frac{1}{2}\right)^2$$

$y = 2$ ...(v)

Putting the value of y in (iv) we get

$x = 3 - y = 3 - 2 = 1$

90. Answer (4)

Hint: Elements whose atomic numbers are more than that of uranium are called transuranium elements.

Sol.:

Ion	Radii (pm)
U^{3+}	103
Am^{3+}	99

[BOTANY]

91. Answer (2)

Hint: Mendel started his research in 1856.

Sol.: Mendel published his research in the year 1865.

92. Answer (2)

Hint: Sahiwal cows are Indian breeds.

Sol.: Sahiwal cows were developed from wild cows via exploitation and artificial selection of certain desirable traits.

93. Answer (4)

Hint: Honeybee shows haplodiploid sex determination.

Sol.: Klinefelter's syndrome is a male specific disorder and has zero probability in females. Haemophilia in females is rare but not impossible.

Drone in honeybee are developed from unfertilised eggs.

Intensity of α-thalassemia increases depending on how many of the alleles are affected by deletion.

94. Answer (3)

Hint: Law of dominance is not universal.

Sol.: The F_1 heterozygous tall progeny would form gametes T and t in equal proportion. *Antirrhinum* (Snapdragon) shows incomplete dominance for flower colour.

95. Answer (1)

Hint: Chromosomal theory of inheritance was proposed is 1902.

Sol.: Chromosomal theory of inheritance was proposed by Sutton.

96. Answer (3)

Hint: Sanger developed the technique to determine sequence of nucleotides in nucleic acid.

Sol.: Sanger developed the technique to determine sequence of amino acids in protein.

97. Answer (2)

Hint: Out of 64 genetic codons, three does not code for any amino acids.

Sol.: The genetic codons UAA, UAG and UGA are stop codons. UAU codes for tyrosine.

98. Answer (4)

Hint: There are total 64 genetic codons.

Sol.: Minimum number of codons required is one per amino acid. So for 20 amino acids, 20 codons are required.

99. Answer (3)

Hint: Down's syndrome occurs due to trisomy of chromosome 21.

Sol.:

(A) An individual with Turner's syndrome-1 X chromosome

(B) Female grasshopper-2 X chromosomes

(C) A female with Down's syndrome-2 X chromosomes

(D) An individual with Klinefelter's syndrome-2 X chromosomes

100. Answer (1)

Hint: 16S rRNA is found in 30S subunit of ribosome.

Sol.: 16S rRNA is found in 70S ribosomes. It is not formed by RNA polymerase I.

101. Answer (1)

Hint: A nucleosome consists of histones and DNA.

Sol.: Prokaryotes lack histones as well as nucleosomes.

102. Answer (1)

Hint: Nuclein refers to DNA.

Sol.: Nuclein (DNA) and Nucleoid (region of DNA in prokaryotes) can be present in *E. coli* together.

103. Answer (2)

Hint: *i*-gene forms mRNA leading to synthesis of repressor.

Sol.: Structural genes -3 (*lac* z, *lac* y, *lac a*)

mRNA-2 (from structural genes and *i* gene)

Enzymes-3 (β-galactosidase, permease and transacetylase)

104. Answer (1)

Hint: Sequencing the whole set of genome and later assigning different regions in the sequence with functions is sequence annotation.

Sol.: Expressed Sequence Tags (ESTs) are focused on identifying all the genes that are expressed as RNA.

105. Answer (4)

Hint: Phenotypic categories can be determined by the formula 2^{n}.

Sol.: The parent TtRr would produce 4 types of gametes leading to 4 different phenotypes in the progeny.

106. Answer (3)

Hint: Formation of gamete and zygote can be understood by a Punnett square.

Sol.:

Discovery of X-body in insects	–	Henking
Graphical representation to show all possible genotypes in offspring	–	Reginald C. Punnett
Experimental verification of chromosomal theory of inheritance	–	T.H. Morgan
Mapped the position of genes on chromosome	–	Alfred Strutevant

107. Answer (1)

Hint: The dihybrid genotypic ratio of F_2 generation is 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1.

Sol.: The F_2 genotypic ratio of a dihybrid cross can be written as following. The progeny that is homozygous for one and heterozygous for the other character are also shown.

RRYY : RrYY : rrYY : RRYy : RrYy : rrYy : RRyy : Rryy : rryy

1 : [2] 1 : [2] 4 : [2] 1 : [2] : 1

$\frac{8}{16} = \frac{1}{2}$

108. Answer (3)

Hint: Independent segregation of two non-allelic genes is called independent assortment.

Sol.: Independent assortment only occurs when two genes are located on separate pairs of homologous chromosomes.

109. Answer (1)

Hint: 0.1% of human genome is unique to each human being.

Sol.:

Genes discovered post HGP that have yet unknown functions	–	More than 50 per cent
Genomic content that codes for protein	–	Less than 2 per cent
Nucleotide sequence identical across all humans	–	More than 99 per cent
Base sequence that may vary in the genomes of two people	–	Less than 0.1 per cent

110. Answer (2)

Hint: Severo Ochoa enzyme is also called polynucleotide phosphorylase.

Sol.: 23S rRNA is a catalytic RNA involved in peptide bond formation, also known as peptidyl transferase.

111. Answer (3)

Hint: The anticodon can form hydrogen bonds with the codon.

Sol.: The 3′ end of tRNA is amino acid acceptor end.

112. Answer (2)

Hint: The number of phosphodiester linkages is equal to the number of nitrogenous bases present in circular DNA.

Sol.: The genome of $\phi \times 174$ phase is a single stranded circular DNA consisting of 5386 nucleotides.

113. Answer (1)

Hint: In case of X-linked recessive disorders, daughters must have affected father.

Sol.: Since individual I-2 and II-3 are affected and have different genotypes, the disorder can only be a dominant disorder. The probability of III-1 being homozygous for myotonic dystrophy is zero because she is affected but has an unaffected father.

114. Answer (1)

Hint: Humans and *Drosophila* are similar in terms of sex determination.

Sol.: Chickens exhibit female heterogamety.

115. Answer (3)

Hint: The male grasshopper contains one less chromosome as compared to the female.

Sol.: Male grasshopper lacks Y chromosome. Y chromosome is found in male *Drosophila*.

116. Answer (1)

Hint: In sickle cell anemia, the mutation changes the 6th codon of mRNA from GAG to GUG.

Sol.: In sickle cell anemia, the mutation causes the Glutamic acid on the 6th position of beta globin chain to be replaced by Valine.

117. Answer (1)

Hint: Axial flower position is considered dominant.

Sol.: Since the flower position determines pod position and terminal flowers are recessive in nature, the probability of having axial green pods in the same plant post fertilisation would be 1.

118. Answer (2)

Hint: Frederick Griffith discovered transformation.

Sol.: Avery, MacLeod and McCarty proved DNA was the transforming principle.

119. Answer (1)

Hint: UGA acts as a non-sense codon.

Sol.: AUG plays an important role as the initiation codon. It also codes for methionine.

120. Answer (1)

Hint: Prokaryotic genes are mostly polycistronic and are non-split in nature.

Sol.: A DNA sequence coding for tRNA, rRNA or polypeptide can define a gene. Most eukaryotic genes are monocistronic and are split genes. Reading frame of mRNA is finally determined post splicing.

121. Answer (1)

Hint: A complete turn of DNA contains 10 bp.

Sol.: If a stretch of DNA contains 150 nucleotide pair,

The total number of turns cannot exceed

$$= \frac{150}{10} = 15 \text{ turns}$$

122. Answer (3)

Hint: Tritiated thymidine helped Taylor to prove the DNA replication model in *Vicia faba*.

Sol.: Deoxyribose sugar is present in DNA.

123. Answer (1)

Hint: One codon codes for one amino acid only.

Sol.: The sequence of codons in mRNA can reliably be determined only if the polypeptide is composed of non-degenerate amino acids.

124. Answer (2)

Hint: Differences in DNA of members of a population at the level of 1 base, are called SNPs or Snips.

Sol.: Allelic sequence variation has traditionally been described as a DNA polymorphism if more than one variant/allele at a locus occurs in human population with a frequency greater than 0.01.

125. Answer (1)

Hint: H. G. Khorana developed the chemical method for synthesising RNA molecules.

Sol.: The technique of DNA Fingerprinting was initially developed by Alec Jeffreys.

126. Answer (2)

Hint: DNA replication occurs in a semiconservative manner.

Sol.: *E. coli* generation time is 20 minutes. After 2 generations (40 minutes), a heavy DNA containing cell, would produce four daughter cells, out of which two will contain light DNA and two would possess hybrid DNA.

127. Answer (1)

Hint: An allele of each pair must be present in gametes.

Sol.: The allele Y does not exist in the genotype of the plant, so it cannot be present in the gamete as well.

128. Answer (3)

Hint: Skin colour becomes darker with increase in number of dominant alleles present in individuals.

Sol.: Three dominant alleles in the genotype lead to intermediate skin colour. Increase in number of dominant alleles by one, would make skin colour slightly darker than the intermediate skin colour.

129. Answer (1)

Hint: Process C is reverse transcription.

Sol.: The relationship between template (mRNA) and product (polypeptide) of translation cannot be explained by using complementarity.

130. Answer (1)

Hint: Humans show XY type sex determination.

Sol.: In grasshopper (XX-XO type), maleness is determined by absence of X chromosome. In honeybee, sex is determined by overall ploidy level of the organism.

131. Answer (3)

Hint: G. Gamow argued that the codon must be composed of a combination of nucleotides.

Sol.: George Gamow suggested that in order to code for all the 20 amino acids, the code should be made up of three nucleotides.

132. Answer (1)

Hint: Individuals with Turner's syndrome are sterile.

Sol.: If both parents are affected and heterozygous for myotonic dystrophy, there is a 25% chance of unaffected progeny. People affected by Down's syndrome can form gametes with normal number of autosomes.

133. Answer (4)

Hint: Both thalassemia and sickle-cell anaemia are recessive gene disorders.

Sol.: Thalassemia is quantitative and sickle-cell anaemia is qualitative problem related to globin molecules.

134. Answer (4)

Hint: Accumulation of ketone bodies in the body occurs in phenylketonuria.

Sol.: Accumulation of phenylpyruvic acid in brain is observed in phenylketonuria.

135. Answer (3)

Hint: Eye colour is an X-linked character in *Drosophila*.

Sol.: Only dominant traits such as brown body colour in *Drosophila* can be expressed in both homozygous as well as heterozygous conditions.

[ZOOLOGY]

136. Answer (4)

Hint: Related to natural selection.

Sol.: According to Charles Darwin, the organic evolution is due to descent with modification of organisms.

137. Answer (2)

Hint: CuT is an example of copper releasing IUD.

Sol.: IUDs do not inhibit oogenesis and ovulation. Contraceptives are not regular requirement for the maintenance of reproductive health.

138. Answer (4)

Hint: Ramification of genetic drift.

Sol.: The scenario described, where a population is drastically and randomly reduced by a natural

calamity (volcanic eruption), is a classic example of the bottleneck effect. The bottleneck effect is a form of genetic drift where chance events cause a severe reduction in population size, and the surviving individuals have a gene pool that may not be representative of the original population's diversity. This leads to a significant loss of genetic variation.

139. Answer (3)

Hint: Results due to branching.

Sol.: Among the stories of evolution of individual species, the story of evolution of modern man most interesting and appears to parallel evolution of human brain and language.

Human evolution is viewed as a very slow, gradual process occurring over millions of years, leading eventually to a single surviving species (*Homo sapiens*).

140. Answer (1)

Hint: Vasectomy

Sol.: Surgical methods, also called sterilisation, are generally advised for the male/female partner as a terminal method to prevent any more pregnancies. Sterilisation procedure in the male is called 'vasectomy'. In vasectomy, a small part of the vas deferens is removed or tied up through a small incision on the scrotum.

141. Answer (4)

Hint: A type of IUD.

Sol.: Diaphragms, cervical caps and vaults are barriers made of rubber that are inserted into the female reproductive tract to cover the cervix during coitus. Condoms are barriers made of thin rubber/latex sheath that are used to cover the penis in the male or vagina and cervix in the female.

142. Answer (3)

Hint: Lived in East African grasslands.

Sol.: *Ramapithecus* was more man-like while *Dryopithecus* was more ape-like. Two mya, *Australopithecus* probably lived in East African grasslands. Evidence shows they hunted with stone weapons but essentially ate fruits. The Neanderthal man with a brain size of 1400 cc lived in near east and central Asia between 1,00,000-40,000 years back.

143. Answer (2)

Hint: Prevalent during Cenozoic era.

Sol.: Conifers were descendants of Psilophyton. Lichens can be used as industrial pollution indicators.

144. Answer (1)

Hint: Exclude the event that took place in 1951.

Sol. Government of India legalised MTP in 1971 with some strict conditions to avoid its misuse. Our population which was approximately 350 million at the time of our independence reached close to the billion marks by 2000 and crossed 1.2 billion in May 2011. India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal. These programmes called 'family planning' were initiated in 1951 and were periodically assessed over the past decades.

145. Answer (3)

Hint: Venereal disease is just another term for STI.

Sol.: One could be free of STI by following the simple principles given below (i) Avoid sex with unknown partners/multiple partners. (ii) Always try to use condoms during coitus. (iii) In case of doubt, one should go to a qualified doctor for early detection and get complete treatment, if diagnosed with infection. Trichomoniasis is caused by a protozoan, *Trichomonas vaginalis*.

146. Answer (2)

Hint: Included under test tube baby programme

Sol.: In test tube baby programme, ova from the wife/donor (female) and sperms from the husband/donor (male) are collected and are induced to form zygote under simulated conditions in the laboratory. The zygote or early embryos (with upto 8 blastomeres) could then be transferred into the fallopian tube (ZIFT-Zygote Intra Fallopian Transfer) and embryos with more than 8 blastomeres, into the uterus (IUT-Intra uterine transfer), to complete its further development.

147. Answer (3)

Hint: True for reptiles and birds

Sol.: The amphibians evolved into reptiles. They lay the shelled eggs which do not dry up in sun unlike those of amphibians.

148. Answer (3)

Hint: Exclude the theory which suggests that Earth is about 4000 years old.

Sol.: Louis Pasteur by careful experimentation demonstrated that life comes only from pre-existing life. He showed that in pre-sterilised flasks, life did not come from killed yeast while in another flask open to air, new living organisms arose from killed yeast. Spontaneous generation theory was dismissed once and for all. However, this did not answer how the first life form came on Earth.

149. Answer (2)

Hint.: Exclude the one who gave the theory of inheritance of acquired character.

Sol.: It is possible that the work of Thomas Malthus on populations influenced Darwin. Natural selection is based on certain observations which are factual. Lamarck proposed the " theory of use and disuse of organs", which is a different evolutionary mechanism. Hugo de Vries is associated with the mutation theory, suggesting evolution occurs through sudden, large changes (saltation).

150. Answer (3)

Hint: The Neanderthal man buried their dead.

Sol.: During ice age, between 75,000-10,000 years ago modern *Homo sapiens* arose. Pre-historic cave art developed about, 18,000 years ago. One such cave paintings by pre-historic humans can be seen at Bhimbetka rock shelter in Raisen district of Madhya Pradesh. Agriculture came around 10,000 years back and human settlements started.

151. Answer (1)

Hint: Basis of lactational amenorrhea

Sol.: Breast feeding stimulates prolactin secretion and evidence suggests that prolactin inhibits GnRH secretion and thus inhibits the secretions of gonadotropins.

152. Answer (2)

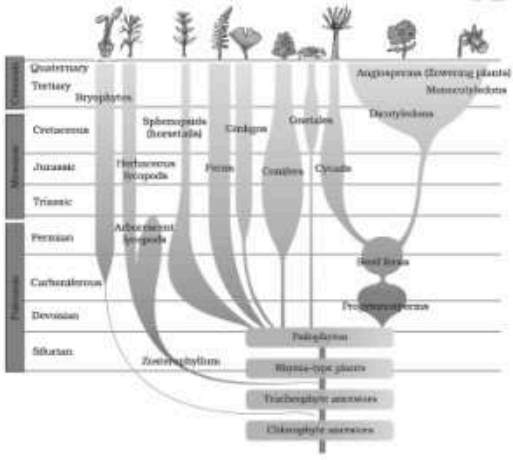
Hint: Decrease in MMR indicates improved maternal health services.

Sol.: A rapid decline in death rate, maternal mortality rate (MMR) and infant mortality rate (IMR) as well as an increase in number of people in reproducible age are probable reasons for population explosion.

153. Answer (2)

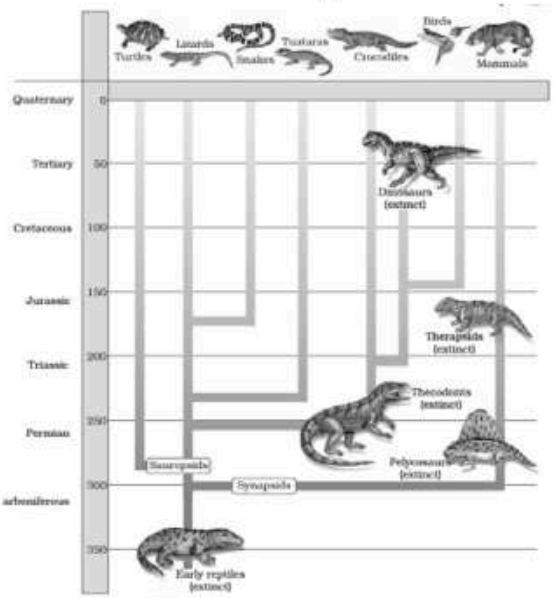
Hint: Exclude the descendants of psilophyton.

Sol.:



154. Answer (2)

Hint: Exclude the one that gave rise to dinosaurs



155. Answer (2)

Hint: Exclude assortative mating.

Sol. Five factors are known to affect Hardy Weinberg equilibrium. These are gene migration or gene flow, genetic drift, mutation, genetic recombination and natural selection.

156. Answer (4)

Hint: True for implants

Sol.: Reproductive Tract Infections (RTIs) specifically affect reproductive organs. However, some sexually transmitted infections (STIs) like AIDS and Hepatitis-B primarily affect the immune system or liver. MTP is considered safe during the first trimester. The risk to the mother increases significantly during the second trimester. IUDs inhibit implantation but not ovulation and insemination.

157. Answer (3)

Hint: Copper releasing IUD

Sol.: Saheli is a non-steroidal oral contraceptive pill developed by CDRI, Lucknow. It is a "once a week" pill with very few side effects. Cu7 is a copper releasing IUD, not hormone releasing IUD. It works by releasing copper ions that suppresses sperm motility and their fertilizing capacity Hormone releasing IUDs include Progestasert and LNG-20. Vasectomy and tubectomy are sterilization or terminal methods. They are highly effective but their reversibility is very poor.

158. Answer (3)

Hint: Happened around 2000 mya

Sol. The first cellular forms of life appeared about 2000 million years ago. Fossils of *Homo erectus* were discovered in Java in 1891. Sea weeds and a few primitive plants existed probably around 320 mya.

159. Answer (1)

Hint: Cranial capacity of neanderthal man was 1400 cc.

Sol. Cranial capacity of *Homo habilis* – 650 to 800 cc.

Cranial capacity of *Australopithecus* – 500 cc

Cranial capacity of *Homo erectus* – 900 cc

160. Answer (2)

Hint: Exclude the techniques that involve *in-vitro* fertilisation.

Sol.: **ZIFT (Zygote Intra-Fallopian Transfer):** Involves *in-vitro* fertilisation where the zygote is later transferred into the fallopian tube.

IUT (Intra-Uterine Transfer): Involves transferring of an embryo (formed *in-vitro*) into the uterus.

ICSI (Intra-Cytoplasmic Sperm Injection): A specialized *in-vitro* procedure where a single sperm is directly injected into an ovum in a laboratory.

161. Answer (3)

Hint: Biogenetic law suggests that ontogeny repeats phylogeny.

Sol.: Embryological support for evolution was proposed by Ernst Heckel based upon the observation of certain features during embryonic stage common to all vertebrates that are absent in adult. For example, the embryos of all vertebrates including human develop a row of vestigial gill slit just behind the head but it is a functional organ only in fish and not found in any other adult vertebrates. However, this proposal was disapproved on careful study performed by Karl Ernst von Baer. He noted that embryos never pass through the adult stages of other animals.

162. Answer (3)

Hint: Different structures evolving for the same function and hence having similarity.

Sol.: Analogous structures are the result of convergent evolution-different structures evolving for the same function and hence having similarity. Other examples of analogy are the eye of the *Octopus* and of mammals. It is the similar habitat that has resulted in selection of similar adaptive features in different groups of organisms but towards the same function.

163. Answer (3)

Hint: Homology is the result of divergent evolution

Sol.: Homology is based on divergent evolution whereas, analogy refers to a situation exactly opposite. Wings of butterfly and of birds look alike. They are not anatomically similar structures though they perform similar functions. Hence, analogous structures are the result of convergent evolution.

164. Answer (4)

Hint: Example of convergent evolution

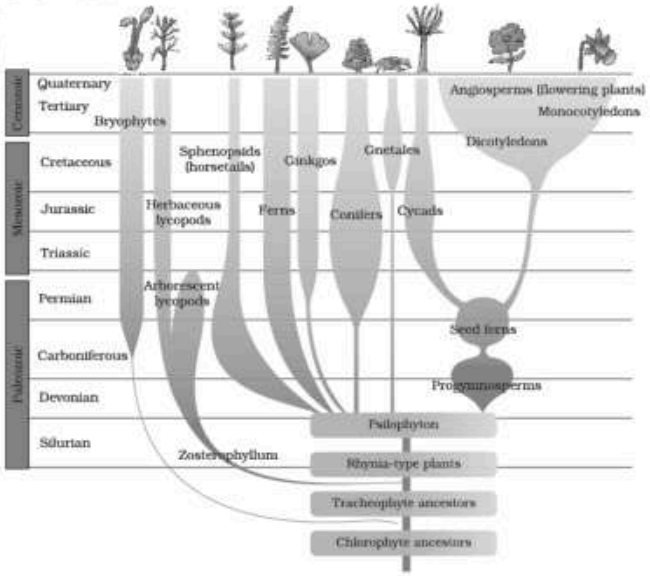
Sol.: The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called adaptive radiation. Darwin's finches represent one of the best example of this phenomenon. Another example is Australian marsupials.

When more than one adaptive radiation appeared to have occurred in an isolated geographical area (representing different habitats), one can call this convergent evolution.

165. Answer (2)

Hint: Progymnosperms and gnetales share the same ancestor.

Sol.:



166. Answer (3)

Hint: Biggest dinosaur

Sol.: About 65 mya, the dinosaurs suddenly disappeared from the Earth. *Tyrannosaurus rex* was about 20 feet in height and had huge fearsome dagger-like teeth. The first non-cellular forms of life could have originated 3 billion years back.

167. Answer (1)

Hint: Heterozygote frequency = 2pq

Sol.: If 320 individuals out of 2000 show the recessive phenotype:

$q^2 = 320/2000 = 0.16$

$q = 0.4$

According to the Hardy Weinberg equation:

$p + q = 1 \Rightarrow p = 0.6$

Heterozygote frequency = $2pq = 2 \times 0.6 \times 0.4 = 0.48$

168. Answer (3)

Hint: Primitive atmosphere was reducing.

Sol. In 1953, S.L. Miller, an American scientist created conditions in a laboratory scale. He created electric discharge in a closed flask containing CH_4, H_2, NH_3 and water vapour at 800°C. He observed formation of amino acids. In similar experiments other observed, formation of sugars, nitrogenous bases, pigment and fats.

169. Answer (2)

Hint: Natural methods of contraception

Sol. Natural methods work on the principle of avoiding chances of ovum and sperms meeting. They do not involve any external devices, chemical, or hormones. IUDs such as Multiload 375, LNG-20 primarily increase sperm phagocytosis or release ions/hormones to prevent implantation. Barrier methods prevent the meeting of gametes; they are mechanical devices (made of rubber/latex) and often used with spermicidal chemicals, unlike natural methods.

Hormonal methods work by inhibiting ovulation and altering the uterine lining.

170. Answer (2)

Hint: Opinion of one medical practitioner is required before 12th week.

Sol. According to the MTP (Amendment) Act, 2017, a pregnancy may be terminated on certain grounds within the first 12 weeks of pregnancy on the opinion of one registered medical practitioner. If the pregnancy has lasted more than 12 weeks, but fewer than 24 weeks, two registered medical practitioners must be of the opinion, formed in good faith, that the required ground exist.

171. Answer (4)

Hint: False for viral STIs

Sol. Infections or diseases which are transmitted *via* sexual intercourse are collectively called sexually transmitted infections (STIs) or venereal diseases. Except for hepatitis-B, genital herpes and HIV infections other STIs are completely curable if detected early and treated properly.

172. Answer (3)

Hint: Ovulation takes place on 14th day.

Sol. Periodic abstinence is an example of natural method of contraception. It is the method in which the couples avoid or abstain from coitus from day 10 to 17 of the 28 days menstrual cycle when ovulation could be expected. As chances of fertilisation are very high during this period, it is called the fertile period. Thus, by abstaining from coitus during this period, conception could be prevented.

173. Answer (1)

Hint: Same mode of action exhibited by implants.

Sol. Administration of progestogens or progestogen-estrogen combination or IUD within 72 hours of coitus have been found to be very effective as emergency contraceptives as they could be used to avoid possible pregnancy due to rape or casual unprotected intercourse.

174. Answer (3)

Hint: True for condoms

Sol.: In barrier methods, ovum and sperms are prevented from physically meeting with the help of barriers. Condoms are barriers made of thin latex/rubber sheath that are used to cover the penis in the male or vagina and cervix in the female reproductive tract. Use of condoms has increased in recent years due to its additional benefit of protecting the user from contracting STIs.

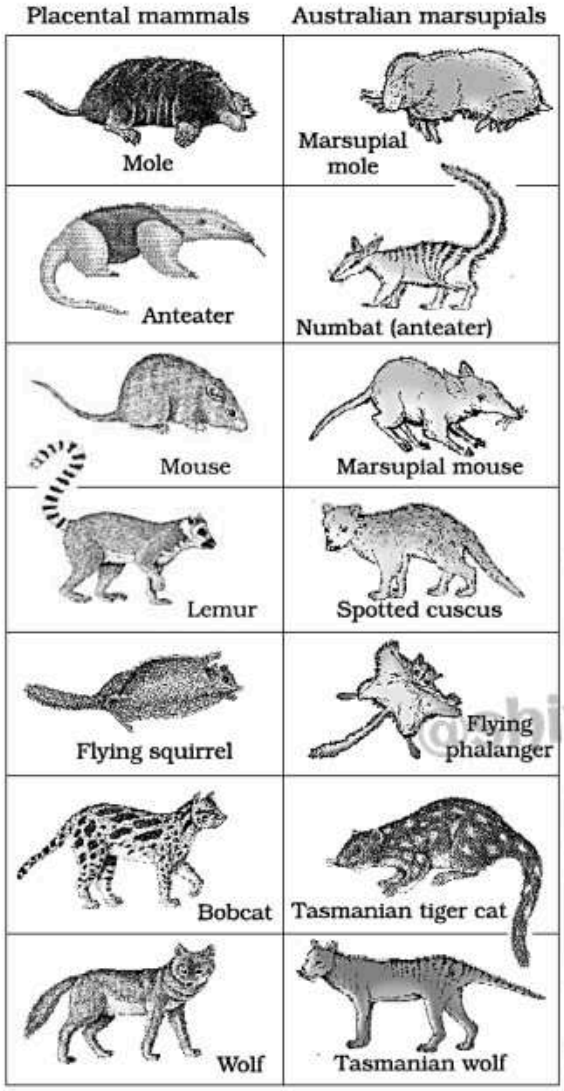
Cu7 is a copper releasing IUD.

In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse the fetal cells.

175. Answer (1)

Hint: Include the organism that shows convergent evolution with flying phalanger.

Sol.:



176. Answer (2)

Hint: Eliminate the concept given by Hugo de Vries.

Sol.: A French naturalist Lamarck had said that evolution of life forms had occurred but driven by use and disuse of organs. He gave the example of Giraffes who in an attempt to forage leaves on tall trees had to adapt by elongation of their necks. As they passed on this acquired character of elongated neck to succeeding generations, Giraffes, slowly, over the years, came to acquire long necks. Nobody believes this conjecture any more.

177. Answer (2)

Hint: White winged moths did not survive.

Sol.: During post-industrialisation period, the tree trunks became dark due to industrial smoke and soots. Under this condition, the white-winged moth did not survive due to predators, dark-winged or melanised moth survived. Before industrialisation set in, thick growth of almost white-coloured lichen covered the trees-in that background the white winged moths survived but the dark-coloured moth were picked out by predators.

178. Answer (3)

Hint: Identify a copper releasing IUD

Sol.: Progestasert and LNG-20 are hormone releasing IUDs that make the uterus unsuitable for implantation and the cervix hostile to the sperms. Progestogens alone or in combination with estrogen can be used by females as injections or implants under the skin.

Multiload-375 is a copper releasing IUD.

179. Answer (4)

Hint: Exclude the viral diseases

Trichomoniasis	–	Protozoal STI caused by *Trichomonas vaginalis*
AIDS	–	Viral STI caused by Human Immunodeficiency Virus
Hepatitis–B	–	Viral STI caused by Hepatitis-B Virus
Chlamydiasis	–	Bacterial STI caused by *Chlamydia trachomatis*

180. Answer (2)

Hint: Cause for population explosion

Sol.: Counselling and creating awareness among people about reproductive organs, adolescence and associated changes, safe and hygienic sexual practices, STIs including AIDS, *etc.*, are the primary steps towards reproductive health. Providing medical facilities and care to the problems like menstrual irregularities, pregnancy related aspects, delivery, medical termination of pregnancy, birth control, infertility, post-natal child and maternal management are important aspects of the Reproductive and Child Health Care programmes.

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